QUESTION ANALYSIS

EEE421

Solid State Devices

Question Analysis

Sessions 2021–22 to 2018–19 *4 papers, 109 questions, 148 formulas*

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1. At a glance:

This analysis covers **4 papers** (2021-22, 2020-21, 2019-20, 2018-19), with 109 questions and 566 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Semiconductor in equilibrium	130	23 %	32	Carrier concentrations (56); Compensation (19); Ionization (17)
Excess carriers	78	14 %	13	Steady state (29); Ambipolar transport (14); Dielectric relaxation (7)
Carrier transport	75	13 %	12	Hall effect (25); Mobility (25); Einstein relation (11)
PN junction	72	13 %	12	Depletion width (25); Ideal diode current (15); Biased junction (10)
Quantum basics	58	10 %	13	Fermi probability (45); Density of states (7); Effective mass (3)

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2021-22	2020-21	2019-20	2018-19	Term tests	All
Crystal structure	·	·	**12**	·	·	**12**
Classes of solids	·	·	**6**	·	·	6
Unit cell	·	·	**6**	·	·	6
Quantum basics	**8**	**15**	·	**19**	**16**	**58**
Fermi probability	**8**	**10**	·	**11**	**16**	45
Density of states	·	**5**	·	**2**	·	7
Effective mass	·	·	·	**3**	·	3
Energy bands	·	·	·	**3**	·	3
Semiconductor in equilibrium	**12**	**30**	**25**	**16**	**47**	**130**
Carrier concentrations	**5**	**5**	**6**	**9**	**31**	56
Compensation	**7**	**5**	**6**	·	**1**	19
Ionization	·	**15**	·	·	**2**	17
Fermi level	·	**5**	·	·	**10**	15
Temperature dependence	·	·	**5**	**7**	·	12
Carrier distribution	·	·	**8**	·	·	8
Dopants	·	·	·	·	**2**	2
Degenerate	·	·	·	·	**1**	1
Carrier transport	·	**25**	**27**	**20**	**3**	**75**
Hall effect	·	**10**	**8**	**7**	·	25
Mobility	·	**10**	**12**	·	**3**	25
Einstein relation	·	**5**	·	**6**	·	11
Diffusion	·	·	·	**7**	·	7
Thermal velocity	·	·	**7**	·	·	7
Excess carriers	·	**20**	**31**	**6**	**21**	**78**

Topic and subtopic	2021-22	2020-21	2019-20	2018-19	Term tests	All
Steady state	·	**10**	**7**	**6**	**6**	29
Ambipolar transport	·	·	**7**	·	**7**	14
Dielectric relaxation	·	·	**7**	·	·	7
Generation and recombination	·	·	·	·	**7**	7
Time and space dependence	·	·	**6**	·	·	6
Minority carriers	·	**5**	·	·	·	5
Carrier pulse	·	**5**	·	·	·	5
Quasi-Fermi levels	·	·	**4**	·	·	4
Excess carriers	·	·	·	·	**1**	1
PN junction	·	**25**	**24**	**20**	**3**	**72**
Depletion width	·	**10**	**12**	·	**3**	25
Ideal diode current	·	·	**8**	**7**	·	15
Biased junction	·	**10**	·	·	·	10
Switching transient	·	·	·	**7**	·	7
Junction capacitance	·	·	·	**6**	·	6
Band diagram	·	**5**	·	·	·	5
Built-in potential	·	·	**4**	·	·	4
Tunnel diode	·	**5**	·	**6**	·	**11**
Operation	·	**5**	·	**6**	·	11
Metal-semiconductor contacts	·	·	**6**	**10**	·	**16**
Schottky and ohmic contacts	·	·	**6**	**10**	·	16
MOS structure	·	**15**	**15**	**24**	·	**54**
Threshold voltage	·	·	**12**	**6**	·	18
Band diagrams	·	**5**	·	**8**	·	13
Body effect	·	·	·	**5**	·	5
C-V characteristics	·	·	·	**5**	·	5
Double gate	·	**5**	·	·	·	5
Inversion	·	**5**	·	·	·	5
MOSFET advantages	·	·	**3**	·	·	3
Bipolar transistor	·	**10**	**10**	**27**	·	**47**
Operating modes	·	**5**	**6**	·	·	11
Emitter efficiency	·	·	**4**	**6**	·	10
Current crowding	·	·	·	**7**	·	7
Amplification	·	·	·	**6**	·	6
Base width modulation	·	·	·	**5**	·	5
Minority distribution	·	**5**	·	·	·	5
Base current	·	·	·	**3**	·	3
JFET	·	·	·	**13**	·	**13**
Pinch-off	·	·	·	**7**	·	7
Frequency response	·	·	·	**6**	·	6

3. Priority list:

Must know: Semiconductor in equilibrium (130), Quantum basics (58)

Likely: Excess carriers (78), Carrier transport (75), PN junction (72), MOS structure (54), Bipolar transistor (47), Metal-semiconductor contacts (16), Tunnel diode (11)

Less often: JFET (13), Crystal structure (12)

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The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

4. Topic pages:

4.1. Crystal structure:

12 marks, 2 questions, in 1 of 4 papers. Book: Neamen Ch. 1.

Most often theory (1 of 2). Subtopics: Classes of solids; Unit cell.

Formulas:

Subtopic	Formula	Equation	Used in
Unit cell (6)	**Atoms per fcc cell**	$n = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$	19-20 1(b)
	Volume density	$\text{density} = \frac{n}{a^3}$	19-20 1(b)

Symbols:

a = lattice constant (cm) ↩

Questions:

No.	Paper	Marks	Type	Question
Classes of solids				
1	19-20 1(a)	6	theory	Describe three classifications of solids-amorphous, polycrystalline, and single crystal.
Unit cell				
2	19-20 1(b)	6	numerical	The lattice constant of a face-centered cubic lattice is 0.425 nm. Determine the (i) effective number of atoms per unit cell (ii) volume density of atoms

4.2. Quantum basics:

58 marks, 13 questions, in 4 of 4 papers. Book: Neamen Ch. 3.

Most often theory (7 of 13). Subtopics: Fermi probability; Density of states; Effective mass; Energy bands.

Formulas:

Subtopic	Formula	Equation	Used in
Fermi probability (45)	**Fermi-Dirac function**	$f_F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$	21-22 7(a), 20-21 8(a), 19-20 7(d), 18-19 1(c), 18-19 2(b)
	Thermal energy	$kT = 0.0259$ eV at 300 K	18-19 2(b)
	Electron concentration	$n_0 = N_c \exp\left[-\frac{E_c - E_F}{kT}\right]$	20-21 8(h), 19-20 7(d)
	Mass-action law	$n_0 p_0 = {n_i}^2 = N_c N_v \exp\left(-\frac{E_g}{kT}\right)$	19-20 7(d)

Subtopic	Formula	Equation	Used in
	Energy for a given probability	$E - E_F = kT \ln\left(\frac{1}{f_F} - 1\right)$	21-22 7(d), 20-21 2(d), 20-21 8(g)
	Ways to place electrons	$W_i = \frac{g_i!}{N_i!\,(g_i - N_i)!}$	21-22 7(a)
	Empty-state probability	$1 - f_F(E) = \frac{1}{1 + \exp\left(\frac{E_F - E}{kT}\right)}$	21-22 7(a)
	Boltzmann approximation	$f_F(E) \approx \exp\left[-\frac{E - E_F}{kT}\right]$	21-22 7(a)
Density of states (7)	**Density of states**	$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c},\ g_v(E) = \frac{4\pi(2m_p^*)^{3/2}}{h^3}\sqrt{E_v - E}$	18-19 7(a)
	Electron concentration	$n_0 = \int_{E_c}^{\infty} g_c(E) f_F(E)\,dE = N_c \exp\left[-\frac{E_c - E_F}{kT}\right]$	18-19 7(a)
	Hole concentration	$p_0 = N_v \exp\left[-\frac{E_F - E_v}{kT}\right]$	18-19 7(a)
	Density of states	$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c}$	20-21 1(c)
	States from E_c to $E_c + kT$	$N = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \cdot \frac{2}{3}(kT)^{3/2}$	20-21 1(c)
Effective mass (3)	**Effective mass**	$\frac{1}{m_n^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}$	18-19 7(c)
Energy bands (3)	**Bloch function**	$\psi(x) = u(x)e^{jkx}$	18-19 7(d)
	Kronig-Penney condition	$P'\frac{\sin\alpha a}{\alpha a} + \cos\alpha a = \cos ka,\ \alpha^2 = \frac{2mE}{\hbar^2}$	18-19 7(d)

Symbols:

$f_F(E)$ = probability that a state at energy E is occupied ↩

E_F = Fermi energy (eV); Fermi level (eV) ↩

kT = thermal energy (0.0259 eV at 300 K) ↩

E = energy of the state (eV); electron energy (J) ↩

E_c = conduction-band edge (eV) ↩

f_F = probability that a state at energy E holds an electron ↩

k = Boltzmann constant (8.62×10^{-5} eV/K); wave number (m^{-1})

T = absolute temperature (K)

n_0 = thermal-equilibrium electron concentration (cm^{-3}) ↩

N_c = effective density of states in the conduction band (2.8×10^{19} cm^{-3} for Si) ↩

n_i = intrinsic carrier concentration (1.5×10^{10} cm^{-3} for Si at 300 K) ↩

E_g = band gap energy (eV), 1.12 eV for Si ↩

W_i = number of ways to place the electrons in level i ↩

g_i = number of states in level i ↩

N_i = number of electrons in level i ↩

$g_c(E)$ = density of states in the conduction band ($cm^{-3}eV^{-1}$) ↩

$g_v(E)$ = density of states in the valence band ($cm^{-3}eV^{-1}$) ↩

m_p^* = hole effective mass (kg) ↩

h = Planck's constant (6.625×10^{-34} J s) ↩

E_v = valence-band edge (eV) ↩

p_0 = thermal-equilibrium hole concentration (cm^{-3}) ↩

N_v = effective density of states in the valence band (cm^{-3}) ↩

g_c = states per unit energy per unit volume in the conduction band ($cm^{-3}eV^{-1}$) ↩

m_n^* = density-of-states effective mass of electrons; electron effective mass

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(kg) ↩

m_0 = free-electron mass (kg)

N = number of states per unit volume in that energy range (cm^{-3}) ↩

$\psi(x)$ = electron wave function in the crystal ↩

$u(x)$ = periodic function, with the period of the lattice ↩

x = position (m)

P' = strength of the potential barriers (dimensionless) ↩

a = width of the well (m) ↩

m = mass of the electron (kg)

$\hbar$ = reduced Planck constant, $h/2\pi$ (J s) ↩

Questions:

No.	Paper	Marks	Type	Question
Fermi probability				
1	21-22 7(a)	3	theory	What is Fermi Dirac distribution?
2	21-22 7(d)	5	numerical repeats 20-21 2(d)	At what energy level above Fermi level (in unit of kT) the probability of finding an electron is 10%?
3	20-21 2(d)	10	numerical	Answer the following questions on Fermi level: (i) At what energy level above Fermi level (in unit of kT) the probability of finding an electron is 10% ? (ii) Find the electron concentration(n_0) for Si at room temperature if Fermi level is 0.5 eV below the conduction band. (at room temperature, $N_c = 2.8 \times 10^{19}$ cm^{-3}).
4	20-21 TT2 8(a)	1	theory	Write down the expression for Fermi Dirac distribution.
5	20-21 TT2 8(g)	5	numerical repeats 20-21 2(d)	At what energy level above Fermi level (in unit of kT) the probability of finding an electron is 10%?
6	20-21 TT2 8(h)	5	numerical repeats 20-21 2(d)	Find the electron concentration (n_o) for Si at room temperature if Fermi level is 0.5 eV below the conduction band. (at room temperature $N_c = 2.8 \times 10^{19}$ cm^{-3})
7	19-20 TT1 7(d)	5	numerical	(i) Calculate the probability of an electron to be occupied at the lowest energy state of the conduction band of silicon when the Fermi energy is 0.3 eV below the conduction band at room temperature. (ii) Calculate the thermal electron concentration n_0. (iii) What is the thermal hole concentration p_0? (Room temperature N_c is 2.8×10^{19} cm^{-3}, n_i is 1.5×10^{10} cm^{-3})
8	18-19 1(c)	5	theory	At absolute zero temperature, what is the maximum energy of an electron in a semiconductor material?
9	18-19 2(b)	6	theory	What is Fermi energy? kT is equal to how many eV at $300k$?
Density of states				
10	20-21 1(c)	5	numerical	Determine the number of quantum states(cm^{-3}) in silicon between E_c and E_c+kT at $T = 300$ K.
11	18-19 7(a)	2	theory	Define the density of states and carrier concentration.
Effective mass				
12	18-19 7(c)	3	theory	What is electron-effective mass? Two possible conduction bands in the E versus k diagram are given in Figure 1. State which band will result in the heavier electron effective mass;state why.
Energy bands				
13	18-19 7(d)	3	theory	Using the necessary theorem explains how band structure is formed of a particular solid-state material.

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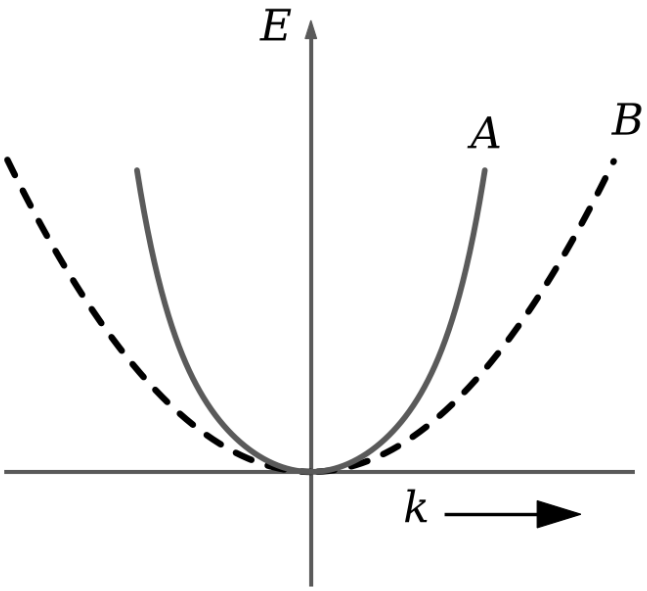


Figure 1: 18-19 7(c): Figure 1: Two possible conduction bands A and B in the E versus k diagram (redrawn from the paper)

4.3. Semiconductor in equilibrium:

130 marks, 32 questions, in 4 of 4 papers. Book: Neamen Ch. 4.

Most often theory (16 of 32). Subtopics: Temperature dependence; Carrier concentrations; Fermi level; Compensation; Carrier distribution; Ionization; Dopants; Degenerate.

Formulas:

Subtopic	Formula	Equation	Used in
Temperature dependence (12)	**Intrinsic concentration**	$n_i{}^2 = N_c N_v \exp\left(-\frac{E_g}{kT}\right)$	18-19 1(a), 18-19 7(b)
	Majority carriers	$p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}$ (and n_0 with $N_d - N_a$)	18-19 1(a), 18-19 1(b), 18-19 8(b)
	Electrons left on donors	$\frac{n_d}{n_d + n_0} = \frac{1}{1 + \frac{N_c}{2N_d}\exp\left[-\frac{E_c - E_d}{kT}\right]}$	20-21 1(d), 20-21 6(b), 20-21 7(a), 19-20 2(c), 18-19 1(a)
	n_i at another temperature	$n_i(T) = n_i(300)\left(\frac{T}{300}\right)^{3/2}\exp\left[-\frac{E_g}{2k}\left(\frac{1}{T} - \frac{1}{300}\right)\right]$	19-20 2(c)
Carrier concentrations (56)	**Electron concentration**	$n_0 = \int_{E_c}^{\infty} g_c(E) f_F(E)\, dE = N_c \exp\left[-\frac{E_c - E_F}{kT}\right]$	18-19 7(b)
	Mass-action law	$n_0 p_0 = n_i^2$	18-19 1(b)
	Mass-action law	$n_0 p_0 = n_i^2 = N_c N_v \exp\left(-\frac{E_g}{kT}\right)$	21-22 7(c), 20-21 1(a), 20-21 7(h), 19-20 2(a), 19-20 7(a)

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Subtopic	Formula	Equation	Used in
	Band gap from n_i	$E_g = kT \ln\left(\frac{N_c N_v}{n_i^2}\right),\ n_i^2 = n_0(n_0 - N_d)$	19-20 7(c)
	Charge neutrality	$n_0 + N_a^- = p_0 + N_d^+$	21-22 7(f), 20-21 3(a), 20-21 8(f), 19-20 2(a), 19-20 7(a)
	Electrons	$n_0 = \frac{N_d - N_a}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}$	20-21 3(a), 20-21 7(g), 20-21 8(f), 19-20 2(a), 19-20 7(a)
	Holes	$p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}$	21-22 7(f), 20-21 3(a), 20-21 8(f), 19-20 2(a), 19-20 7(a)
	Carriers per unit energy	$n_0 = \int_{E_c}^{\infty} g_c(E) f_F(E)\, dE,\ p_0 = \int_{-\infty}^{E_v} g_v(E)[1 - f_F(E)]\, dE$	20-21 7(f)
	From the intrinsic level	$n_0 = n_i \exp\left(\frac{E_F - E_{Fi}}{kT}\right),\ p_0 = n_i \exp\left(-\frac{E_F - E_{Fi}}{kT}\right)$	20-21 7(f)
	Intrinsic at high T	$n_0 = \frac{N_d}{2} + \sqrt{\left(\frac{N_d}{2}\right)^2 + n_i^2(T)},\ n_i^2 \propto T^3 e^{-E_g/kT}$	20-21 7(e)
	Integral used	$\int_0^{\infty} \eta^{1/2} e^{-\eta}\, d\eta = \frac{\sqrt{\pi}}{2}$	21-22 7(c)
	From the intrinsic level	$n_0 = n_i \exp\left[\frac{E_F - E_{Fi}}{kT}\right],\ p_0 = n_i \exp\left[-\frac{E_F - E_{Fi}}{kT}\right]$	21-22 7(c)
	Fermi level, n-type	$E_F - E_{Fi} = kT \ln\frac{n_0}{n_i},\ E_c - E_F = kT \ln\frac{N_c}{n_0}$	21-22 7(c)
	Fermi level, p-type	$E_{Fi} - E_F = kT \ln\frac{p_0}{n_i},\ E_F - E_v = kT \ln\frac{N_v}{p_0}$	21-22 7(c), 21-22 7(f)
Fermi level (15)	Hole concentration	$p_0 = N_v \exp\left[-\frac{E_F - E_v}{kT}\right]$	21-22 7(c), 20-21 1(a), 20-21 7(f), 19-20 3(a), 18-19 8(b)
	Fermi level from the band edge	$E_F - E_v = kT \ln\frac{N_v}{p_0}$	18-19 8(b)
	Fermi level from the intrinsic level	$E_{Fi} - E_F = kT \ln\frac{p_0}{n_i}$	18-19 8(b)
	Fermi level	$E_F - E_{Fi} = kT \ln\frac{n_0}{n_i},\ E_{Fi} - E_F = kT \ln\frac{p_0}{n_i}$	19-20 7(b)
	Fermi level from the band edges	$E_c - E_F = kT \ln\frac{N_c}{n_0},\ E_F - E_v = kT \ln\frac{N_v}{p_0}$	19-20 7(b)

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Subtopic	Formula	Equation	Used in
	Fermi level vs doping	$E_F - E_{Fi} = kT \ln \frac{N_d}{n_i}$ (n-type), $E_{Fi} - E_F = kT \ln \frac{N_a}{n_i}$ (p-type)	20-21 1(a)
	Fermi level from E_v (p-type)	$E_F - E_v = kT \ln \frac{N_v}{N_a}$	20-21 8(c)
Compensation (19)	Amphoteric split	$N_d = x_{Ga} N_{Si}$, $N_a = x_{As} N_{Si}$	21-22 7(f)
	Minority carriers	$n_0 = \frac{n_i^2}{p_0}$	21-22 7(f)
Carrier distribution (8)	**Fermi-Dirac function**	$f_F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$	20-21 1(a), 19-20 3(a)
	Density of states	$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c}$	21-22 7(c), 19-20 3(a)
	Carrier distributions	$n(E) = g_c(E) f_F(E)$, $p(E) = g_v(E)[1 - f_F(E)]$	21-22 7(c), 19-20 3(a)
	Electron concentration	$n_0 = N_c \exp\left[-\frac{E_c - E_F}{kT}\right]$	21-22 7(c), 21-22 7(e), 20-21 7(f), 19-20 3(a)
Ionization (17)	N_c at another temperature	$N_c(T) = N_c(300)\left(\frac{T}{300}\right)^{3/2}$	20-21 1(d)
	Fermi level from E_c	$E_c - E_F = kT \ln \frac{N_c}{n_0}$	20-21 1(d)
	Donor occupation	$f_d = \frac{1}{1 + \frac{1}{2}\exp\left(\frac{E_d - E_F}{kT}\right)}$	20-21 1(d), 20-21 7(b)
	Acceptors still neutral	$\frac{p_a}{p_a + p_0} = \frac{1}{1 + \frac{N_v}{4N_a}\exp\left[-\frac{E_a - E_v}{kT}\right]}$	20-21 7(a)
	Freeze-out estimate	$n_0 \approx \sqrt{\frac{N_c N_d}{2}}\exp\left[-\frac{E_c - E_d}{2kT}\right]$	20-21 1(d)
	Bohr orbit radius	$\frac{r_n}{a_0} = n^2 \epsilon_r \frac{m_0}{m^*}$	21-22 7(b)
	Donor energy (hydrogen model)	$E = -\frac{m^* e^4}{2(n\hbar)^2(4\pi\epsilon)^2}$	21-22 7(b)
	Ionization energy	$E_{ion} = 13.6\text{ eV} \times \frac{m^*/m_0}{\epsilon_r^2}$	21-22 7(b)
	Donor and acceptor	donor: $E_c - E_d$; acceptor: $E_a - E_v$	21-22 7(b), 21-22 7(f)
Dopants (2)	Si as a dopant in GaAs	Si on a Ga site: donor; Si on an As site: acceptor	20-21 7(d), 20-21 8(d)
Degenerate (1)	Degenerate limit	E_F inside a band when $n_0 > N_c$ (or $p_0 > N_v$)	20-21 8(e)

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Symbols:

n_i = intrinsic carrier concentration (1.5×10^{10} cm^{-3} for Si at 300 K) ↩

E_g = band gap energy (eV), 1.12 eV for Si ↩

N_a = acceptor concentration (cm^{-3}) ↩

N_d = donor concentration (cm^{-3}) ↩

n_d = electrons still bound to donor atoms (cm^{-3}); electrons still in donor states (cm^{-3}) ↩

E_d = donor energy level (eV) ↩

$n_i(300)$ = intrinsic concentration at 300 K (cm^{-3}) ↩

n_0 = thermal-equilibrium electron concentration (cm^{-3}) ↩

N_c = effective density of states in the conduction band, $2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}$ (cm^{-3}) ↩

N_a^- = ionized acceptor concentration (cm^{-3}) ↩

N_d^+ = ionized donor concentration (cm^{-3}) ↩

$g_v(E)$ = density of states in the valence band: the same with m_p^* and $\sqrt{E_v - E}$ ↩

η = energy above the band edge in units of kT, $(E - E_c)/kT$ ↩

E_{Fi} = intrinsic Fermi level, near midgap (eV) ↩

$n_i(T)$ = intrinsic concentration at temperature T (cm^{-3}) ↩

p_0 = thermal-equilibrium hole concentration (cm^{-3}) ↩

N_v = effective density of states in the valence band, $2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}$ (cm^{-3}) ↩

E_v = valence-band edge (eV) ↩

N_{Si} = silicon atoms added to the GaAs (cm^{-3}) ↩

x_{Ga} = fraction of Si on Ga sites (donors) ↩

x_{As} = fraction of Si on As sites (acceptors) ↩

$f_F(E)$ = probability that a state at energy E is occupied ↩

E = energy of the state (eV) ↩

E_F = Fermi energy (eV); Fermi level (eV) ↩

kT = thermal energy (0.0259 eV at 300 K) ↩

E_c = conduction-band edge (eV) ↩

f_F = probability that a state at energy E holds an electron ↩

$g_c(E)$ = density of states in the conduction band (cm^{-3}eV^{-1}) ↩

m_n^* = electron effective mass (kg) ↩

h = Planck's constant (6.625×10^{-34} J s) ↩

m_p^* = hole effective mass (kg)

$n(E)$ = electrons per unit energy in the conduction band (cm^{-3}eV^{-1}) ↩

$p(E)$ = holes per unit energy in the valence band (cm^{-3}eV^{-1}) ↩

T = absolute temperature (K) ↩

$N_c(300)$ = effective density of states at 300 K (cm^{-3}) ↩

f_d = probability that a donor state holds an electron ↩

p_a = holes still in acceptor states (cm^{-3}) ↩

E_a = acceptor energy level (eV) ↩

g = degeneracy factor of the acceptor level (4)

r_n = radius of the n-th orbit (Å) ↩

a_0 = Bohr radius (0.53 Å) ↩

n = orbit number (1 for the ground state) ↩

ϵ_r = relative permittivity (11.7 for Si) ↩

m_0 = free-electron mass (kg) ↩

m^* = conductivity effective mass (kg) ↩

e = magnitude of the electron charge (1.6×10^{-19} C) ↩

$\hbar$ = reduced Planck constant, $h/2\pi$ (J s) ↩

ϵ = permittivity of the semiconductor, $\epsilon_r\epsilon_0$ (F/m) ↩

E_{ion} = donor ionization energy (eV) ↩

Questions:

No.	Paper	Marks	Type	Question
Temperature dependence				
1	19-20 2(c)	5	theory repeats 18-19 1(a)	Using proper graph, explain how carrier concentration varies with temperature.
2	18-19 1(a)	7	theory	Explain the temperature dependance *[sic: should be "dependence"]* of carrier concentration of an extrinsic semiconductor with the help of graph.
Carrier concentrations				
3	21-22 7(c)	5	derivation repeats 20-21 7(f)	Derive the equations of equilibrium distribution of electron and holes for extrinsic semiconductor material.
4	21-22 7(e)	0	numerical repeats 20-21 2(d)	Find the electron concentration (n_o) for Si at room temperature if Fermi level is 0.5 eV below the conduction band. (at room temperature N_c =2.8 × 10^{19} cm^{-3})
5	20-21 3(a)	5	derivation repeats 19-20 7(a)	Starting from charge neutrality condition $n_0 + N_a^- = p_0 + N_d^+$, derive the expressions for n_0 and p_0 in terms of N_a, N_d, and n_i.

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No.	Paper	Marks	Type	Question
6	20-21 TT1 7(e)	1	theory	True or false: At high temperature extrinsic semiconductor stars *[sic: should be "starts"]* behaving like intrinsic semiconductor?
7	20-21 TT1 7(f)	5	derivation repeats 20-21 3(a)	Derive the equations of equilibrium distribution of electron and holes for extrinsic semiconductor material.
8	20-21 TT1 7(g)	5	numerical	What will be the majority carrier concentration of a n-type semiconductor at 300K with $N_d = 10^6$ cm^{-3} *[sic: should be* 10^{16} cm^{-3}*]* and carrier concentration of $n_i = 1.5 \times 10^{10}$ cm^{-3}.
9	20-21 TT1 7(h)	5	numerical repeats 20-21 2(d)	What will be the minority carrier concentration.
10	20-21 TT2 8(f)	5	derivation repeats 20-21 3(a)	Starting from charge neutrality condition $n_o + N^{a-}$ *[sic: should be* N_a^-*]* $= p_o + N_d^+$, derive expressions for n_o and p_o in terms of N_a, N_d, and n_i.
11	19-20 2(a)	6	derivation	Starting from charge neutrality condition, Find thermal equilibrium carrier concentration at compensated semiconductor in terms of N_a, N_d and n_i
12	19-20 TT1 7(a)	5	derivation repeats 19-20 2(a)	Starting from charge neutrality condition, derive expressions for n_0 and p_0 in terms of N_a, N_d, and n_i.
13	19-20 TT1 7(c)	5	numerical	A special semiconductor material is to be "designed". The semiconductor is to be n-type and doped with $1 \times 10^{15} cm^{-3}$ donor atoms. Assume complete ionization and assume $N_a = 0$. The effective density of states functions are given by $N_c = N_v = 1.5 \times 10^{19} cm^{-3}$ and are independent of temperature. A particular semiconductor device fabricated with this material requires the electron concentration to be no greater than $1.01 \times 10^{15} cm^{-3}$ at $T = 400\ K$. What is the minimum value of the bandgap energy?
14	18-19 1(b)	6	numerical	A germanium sample is doped with 10^{15} boron atoms per cm^3. Find the electron concentration. Intrinsic carrier concentration of germanium is $2.5 \times 10^{13}/cm^3$ at $300k$.
15	18-19 7(b)	3	theory	Explain with the help of equations and graphs- why intrinsic carrier concentration (n_i) for a typical semiconductor is high (10^6-10^9/cc) at 300K even though the Fermi Dirac distribution is low at that temperature.
Fermi level				
16	20-21 1(a)	5	theory	Sketch the position of Fermi level in Si at 300 K as a function of doping concentration.
17	20-21 TT2 8(c)	1	diagram	Sketch E_c, E_v, E_{Fi} and E_F for a p type extrinsic semiconductor.
18	19-20 TT1 7(b)	5	theory	With proper equation and graph, explain variation of Fermi energy level in extrinsic semiconductor.
19	18-19 TT2 8(b)	4	numerical	Determine the position of the Fermi level with respect to the valence band energy in p type GaAs at T = 300 K. The doping concentrations are [unclear: N_a] $= 5 \times 10^{19}$ *[sic: should be* 5×10^{16}*]* cm^{-3} and $N_d = 10^{15}$ *[sic: should be* 4×10^{15}*]* cm^{-3}.
Compensation				
20	21-22 7(f)	7	numerical	Silicon atoms, at a concentration of 7×10^{15} cm^{-3}, are added to gallium arsenide. Assume that the silicon atoms act as fully ionized dopant atoms and that 5 percent of the concentration added replaces gallium atoms and 95 percent replaces arsenic atoms. Let T = 300 K and GaAs has an intrinsic concentration of 1.8x10^6 cm $^{-3}$. (i) Determine the donor and acceptor concentration. (ii) Is the material n-type or p-type? (iii) Calculate the electron and hole concentration. (iv) Determine the position of the Fermi level with respect to E_{Fi}.
21	20-21 2(a)	5	theory repeats 19-20 2(d)	Draw energy band diagram of a compensated semiconductor.

No.	Paper	Marks	Type	Question
22	20-21 TT2 8(b)	1	theory	What is compensated semiconductor?
23	19-20 2(d)	6	theory	Sketch and briefly describe energy band diagram of compensated semiconductor.
Carrier distribution				
24	19-20 3(a)	8	theory	Sketch and explain density of states functions, Fermi distribution, and carrier concentration distribution curves of n-type material in a single graph.
Ionization				
25	21-22 7(b)	0	theory	Explain Ionization Energy.
26	20-21 1(d)	10	numerical	A Si sample is doped with $N_d = 1 \times 10^{15}$ cm^{-3} of As atoms. Calculate the electron and hole concentrations and the position of Fermi level position (w.r.t. E_c/E_v) at 300 K, assuming full ionization. Justify the assumption by calculating the probability of doner *[sic: should be donor]* states being occupied by electrons and therefore not ionized (assume, $E_C - E_D = 50$ meV). Repeat the above calculations for $T = 30$ K and comment on the values and assumptions. [$N_c = 2.8 \times 10^{19}$ cm^{-3}, for Si].
27	20-21 6(b)	5	derivation	Briefly explain the terms complete ionization and freeze out. Can we achieve complete ionization in room temperature? Prove your answer.
28	20-21 TT1 7(a)	1	theory	What fraction of donor or acceptor states are ionized at room temperature?
29	20-21 TT1 7(b)	1	theory	What is the probability function of electron occupying the donor state??
Dopants				
30	20-21 TT1 7(d)	1	theory	Can Si be used as a donor? How?
31	20-21 TT2 8(d)	1	theory repeats 20-21 7(d)	Can Si be used as a donor? How?
Degenerate				
32	20-21 TT2 8(e)	1	theory	What is degenerated semiconductor?

4.4. Carrier transport:

75 marks, 12 questions, in 3 of 4 papers. Book: Neamen Ch. 5.

Most often numerical (5 of 12). Subtopics: Hall effect; Einstein relation; Diffusion; Mobility; Thermal velocity.

Formulas:

Subtopic	Formula	Equation	Used in
Hall effect (25)	**Hall voltage**	$V_H = \frac{I_x B_z}{epd}$; $p = \frac{I_x B_z}{edV_H}$, $n = -\frac{I_x B_z}{edV_H}$	18-19 1(d)
	Hall mobility	$\mu_p = \frac{I_x L}{epV_x W d}$	18-19 1(d)
	Hall voltage	$V_H = E_H W$, $V_H = -\frac{I_x B_z}{ned}$	19-20 2(b)
	Hall concentration and mobility	$n = -\frac{I_x B_z}{edV_H}$, $\mu_n = \frac{I_x L}{enV_x W d}$	19-20 2(b)
	Hall voltage	$V_H = \frac{I_x B_z}{epd}$ (p-type), $V_H = -\frac{I_x B_z}{end}$ (n-type)	20-21 5(d)

Subtopic	Formula	Equation	Used in
	Carrier density, Hall	$n = -\frac{I_x B_z}{e d V_H}$	20-21 5(d)
	Hall mobility	$\mu_n = \frac{I_x L}{e n V_x W d}$	20-21 5(d)
Einstein relation (11)	**Einstein relation**	$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{e}$	20-21 2(b), 18-19 2(c)
Diffusion (7)	**Diffusion current**	$J_{n,dif} = eD_n\frac{dn}{dx}$, $J_{p,dif} = -eD_p\frac{dp}{dx}$	18-19 5(a)
	Total current	$J = en\mu_n E + ep\mu_p E + eD_n\frac{dn}{dx} - eD_p\frac{dp}{dx}$	18-19 5(a)
Mobility (25)	**Mass-action law**	$n_0 p_0 = n_i^2$	18-19 8(c)
	Drift and conductivity	$\sigma = e(\mu_n n + \mu_p p)$, $\rho = \frac{1}{\sigma}$	18-19 8(c)
	Mobility	$\mu = \frac{e\tau}{m^*}$, $v_d = \mu E$	19-20 4(a)
	Two scattering mechanisms	$\mu_L \propto T^{-3/2}$, $\mu_I \propto \frac{T^{3/2}}{N_I}$, $\frac{1}{\mu} = \frac{1}{\mu_L} + \frac{1}{\mu_I}$	20-21 4(a), 19-20 4(a), 19-20 3(d)
	Conductivity	$\sigma = e(\mu_n n + \mu_p p)$	19-20 3(d)
	Drift velocity	$v_d = \mu E$	20-21 3(c)
	Mobility	$\mu = \frac{e\tau}{m^*}$	20-21 3(c)
	Conductivity	$\sigma = e(n\mu_n + p\mu_p)$	20-21 3(c)
Thermal velocity (7)	Maxwell-Boltzmann speeds	$f(v) = 4\pi\left(\frac{m}{2\pi kT}\right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2kT}\right)$	19-20 1(c)
	Mean speed	$\bar{v} = \sqrt{\frac{8kT}{\pi m^*}}$	19-20 1(c)
	rms (thermal) speed	$v_{rms} = \sqrt{\frac{3kT}{m^*}}$	19-20 1(c)

Symbols:

V_H = Hall voltage (V) ↩

I_x = current along x (A) ↩

B_z = magnetic flux density along z (T) ↩

d = sample thickness, along B_z (cm)

L = sample length (cm); length along x (cm)

V_x = applied voltage along x (V) ↩

W = sample width, across the Hall field (cm); width along y (cm)

E_H = Hall field (V/cm) ↩

kT/e = thermal voltage (0.0259 V at 300 K)

$J_{n,dif}$ = electron diffusion current density (A/cm^2) ↩

$J_{p,dif}$ = hole diffusion current density (A/cm^2) ↩

D_n = electron diffusion coefficient (cm^2/s) ↩

D_p = hole diffusion coefficient (cm^2/s) ↩

F = particle flux (cm^{-2}s^{-1})

v_{th} = thermal velocity (cm/s)

l = mean free path (cm)

J = total current density (A/cm^2) ↩

E = electric field (V/cm) ↩

σ = conductivity (Ω-cm)$^{-1}$ ↩

ρ = resistivity, $1/\sigma$ (Ω cm) ↩

e = magnitude of the electron charge (1.6×10^{-19} C) ↩

μ_n = electron mobility (cm^2/V s) ↩

μ_p = hole mobility (cm^2/V s) ↩

n = electron concentration (cm^{-3}) ↩

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p = hole concentration (cm^{-3}) ↩

N_I = total ionized impurity concentration, $N_d^+ + N_a^-$ (cm^{-3}) ↩

μ = mobility ($cm^2/V\ s$) ↩

τ = mean time between collisions (s) ↩

v_d = drift velocity (cm/s) ↩

μ_L = mobility limited by lattice (phonon) scattering ↩

μ_I = mobility limited by ionized-impurity scattering ↩

m^* = conductivity effective mass (kg); effective mass, in units of m_0 = free-electron mass ↩

$f(v)$ = fraction of carriers per unit speed ↩

v = carrier speed (m/s) ↩

m = effective mass of the carrier (kg) ↩

k = Boltzmann constant (1.38×10^{-23} J/K)

T = absolute temperature (K) ↩

$\bar{v}$ = mean thermal speed (m/s) ↩

v_{rms} = root-mean-square thermal speed (m/s) ↩

Questions:

No.	Paper	Marks	Type	Question
Hall effect				
1	20-21 5(d)	10	numerical	Current is flowing through an extrinsic semiconductor placed in a magnetic field as shown in Fig. 1. (i) What is the Hall voltage? (ii) What is type of carrier? (iii) What is the carrier concentration? (iv) What is the mobility?
2	19-20 2(b)	8	numerical	Consider silicon at $T = 300$ K. A Hall effect device is fabricated with the following geometry: $d = 5 \times 10^{-3}$ cm, $W = 5 \times 10^{-2}$ cm and $L = 0.50$ cm. The electrical parameters measured are: $I_x = 0.50$ mA, $V_x = 1.25$ V, and $B_z = 650$ *gauss* $= 6.5 \times 10^{-2}$ *tesla*. The Hall field is $E_H = -16.5$ mV cm^{-1}. Determine: (i) the Hall voltage (ii) the conductivity type (iii) the majority carrier concentration and (iv) the majority carrier mobility
3	18-19 1(d)	7	derivation	What is the Hall effect? Derive the expression for finding the carrier concentration of a semiconductor from Hall voltage.
Einstein relation				
4	20-21 2(b)	5	derivation repeats 18-19 2(c)	Derive the Einstein relation relating the mobility and diffusion constant. Also, discuss the significance of the Einstein relation.
5	18-19 2(c)	6	derivation	What is Einstein Relation? Derive the expression.
Diffusion				
6	18-19 5(a)	7	derivation	Describe diffusion process. Derive the expression for diffusion current density.
Mobility				
7	20-21 3(c)	5	theory	Define drift mobility. Explain the mechanism of carrier scattering in semiconductor materials and significance of τ.
8	20-21 4(a)	5	theory repeats 19-20 4(a)	Explain the effects of temperature on carrier mobility inside a semiconductor.
9	19-20 3(d)	6	numerical	Consider a compensated n-type silicon at $T = 300$ K with a conductivity of ρ *[sic: should be σ]* $= 16\ (\Omega - cm)^{-1}$ and an acceptor doping concentration of 10^{17} cm^{-3}. Determine the doner *[sic: should be donor]* concentration and the electron mobility.
10	19-20 4(a)	6	theory	With proper graphs, explain how mobility is affected by temperature and doping concentration.
11	18-19 TT2 8(c)	3	numerical	Silicon at T = 300 K is doped with impurity concentrations of Nd = 5 × 1016 *[sic: should be 10^{16}]* /cm3 and Na = 2 × 1016 *[sic: should be 10^{16}]* /cm3. (a) What are the electron and hole mobilities? (b) Determine the conductivity and resistivity of the [unclear: material].
Thermal velocity				
12	19-20 1(c)	7	numerical	If the free electrons in a semiconductor have effective mass $m_c = 0.1m_o$, where m_o is the mass of the electron in vacuum, what is their average speed at $T = 300K$? If the free holes in the same material have effective mass $m_v = 1.5m_o$, what is their average speed at the same temperature? Assume a Maxwell-Boltzmann distribution of the electron energies.

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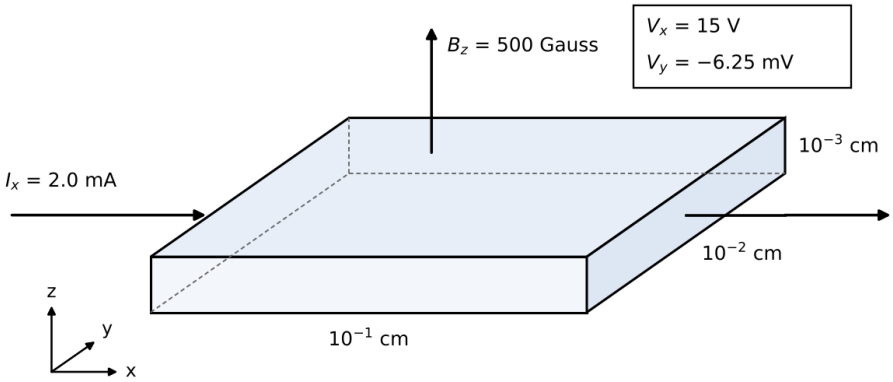


Figure 2: 20-21 5(d): Fig. 1: A bar of an extrinsic semiconductor in a magnetic field (redrawn from the paper)

4.5. Excess carriers:

78 marks, 13 questions, in 3 of 4 papers. Book: Neamen Ch. 6.

Most often derivation (6 of 13). Subtopics: Steady state; Dielectric relaxation; Ambipolar transport; Time and space dependence; Quasi-Fermi levels; Generation and recombination; Minority carriers; Carrier pulse; Excess carriers.

Formulas:

Subtopic	Formula	Equation	Used in
Steady state (29)	**Diffusion length**	$L_p = \sqrt{D_p \tau_{p0}}$, $\delta p(x) = \delta p_0 e^{-x/L_p}$	18-19 5(c)
	Steady state, source at $x = 0$	$\delta n(x) = \delta n(0) e^{-\lvert x\rvert/L_n}$, $L_n = \sqrt{D_n \tau_{n0}}$	19-20 5(a), 19-20 8(a), 19-20 4(d)
	Ambipolar transport	$D' \frac{\partial^2(\delta n)}{\partial x^2} + \mu' E \frac{\partial(\delta n)}{\partial x} + g' - R' = \frac{\partial(\delta n)}{\partial t}$	20-21 3(d)
	Steady-state diffusion	$\delta n(x) = \delta n(0) \exp\left(-\frac{x}{L_n}\right)$, $L_n = \sqrt{D_n \tau_{n0}}$	20-21 3(d)
Dielectric relaxation (7)	**Dielectric relaxation**	$\rho(t) = \rho(0) e^{-t/\tau_d}$, $\tau_d = \frac{\epsilon}{\sigma}$	19-20 3(c)
Ambipolar transport (14)	Current density	$J = en\mu_n E + ep\mu_p E + eD_n \frac{dn}{dx} - eD_p \frac{dp}{dx}$	19-20 4(b)
	Einstein relation	$\frac{\mu_n}{D_n} = \frac{\mu_p}{D_p} = \frac{e}{kT}$	19-20 4(b)
	Continuity	$\frac{\partial p}{\partial t} = -\frac{\partial F_p^+}{\partial x} + g_p - \frac{p}{\tau_{pt}}$	19-20 4(b), 19-20 8(b)
	Ambipolar transport equation	$D' \frac{\partial^2(\delta n)}{\partial x^2} + \mu' E \frac{\partial(\delta n)}{\partial x} + g - R = \frac{\partial(\delta n)}{\partial t}$	19-20 4(b), 19-20 8(b)
	Ambipolar coefficients	$D' = \frac{D_n D_p (n+p)}{D_n n + D_p p}$, $\mu' = \frac{\mu_n \mu_p (p-n)}{\mu_n n + \mu_p p}$	19-20 4(b), 19-20 8(b)

Subtopic	Formula	Equation	Used in
Time and space dependence (6)	**Uniform decay and build-up**	$\delta p(t) = \delta p(0)e^{-t/\tau_{p0}}$, $\delta p(t) = g'\tau_{p0}(1 - e^{-t/\tau_{p0}})$	19-20 4(d)
	Pulse	$\delta p(x,t) = \frac{e^{-t/\tau_{p0}}}{(4\pi D_p t)^{1/2}} \exp\left[-\frac{(x-\mu_p E_0 t)^2}{4D_p t}\right]$	19-20 4(d)
Quasi-Fermi levels (4)	**Quasi-Fermi levels**	$E_{Fn} - E_{Fi} = kT \ln \frac{n_0 + \delta n}{n_i}$, $E_{Fi} - E_{Fp} = kT \ln \frac{p_0 + \delta p}{n_i}$	19-20 5(c)
Generation and recombination (7)	Direct recombination	$\frac{dn}{dt} = \alpha_r(n_i^2 - np)$, $R' = \frac{\delta n}{\tau_{n0}}$	19-20 8(c)
	SRH recombination	$R = \frac{C_n C_p N_t (np - n_i^2)}{C_n(n + n') + C_p(p + p')}$	19-20 8(c)
Minority carriers (5)	**Diffusion current**	$J_n = eD_n \frac{dn}{dx}$, $J_p = -eD_p \frac{dp}{dx}$	20-21 1(b)
	Poisson's equation	$\frac{d^2\phi}{dx^2} = -\frac{\rho(x)}{\epsilon_s} = -\frac{dE}{dx}$	20-21 1(b)
Carrier pulse (5)	n-type, low injection	$D_p \frac{\partial^2(\delta p)}{\partial x^2} - \mu_p E_0 \frac{\partial(\delta p)}{\partial x} - \frac{\delta p}{\tau_{p0}} = \frac{\partial(\delta p)}{\partial t}$	20-21 5(a)
	Moving pulse	$\delta p(x,t) = \frac{N_0}{\sqrt{4\pi D_p t}} \exp\left(-\frac{t}{\tau_{p0}}\right) \exp\left[-\frac{(x-\mu_p E_0 t)^2}{4D_p t}\right]$	20-21 5(a)
Excess carriers (1)	Total and excess carriers	$n = n_0 + \delta n$, $p = p_0 + \delta p$	20-21 7(c)

Symbols:

L_p = hole diffusion length (cm) ↩

τ_{p0} = minority-carrier (hole) lifetime (s); minority-hole lifetime (s) ↩

$\delta p(x)$ = excess hole concentration at x (cm^{-3}) ↩

δp_0 = excess hole concentration at $x = 0$ (cm^{-3}) ↩

x = distance from the source (cm); distance from the injection plane (cm) ↩

L_n = minority-electron diffusion length (cm) ↩

τ_{n0} = minority-carrier (electron) lifetime (s); minority-electron lifetime (s) ↩

D' = ambipolar diffusion coefficient (cm^2/s) ↩

μ' = ambipolar mobility ($\text{cm}^2/\text{V s}$) ↩

g' = excess-carrier generation rate ($\text{cm}^{-3}\text{s}^{-1}$) ↩

R' = excess-carrier recombination rate ($\text{cm}^{-3}\text{s}^{-1}$); recombination rate of the excess carriers ($\text{cm}^{-3}\text{s}^{-1}$) ↩

t = time (s) ↩

ρ = net charge density (C/cm^3) ↩

τ_d = dielectric relaxation time constant (s) ↩

ϵ = permittivity of the semiconductor (F/cm) ↩

D_n = electron diffusion coefficient (cm^2/s) ↩

D_p = hole diffusion coefficient (cm^2/s) ↩

kT/e = thermal voltage (0.0259 V at 300 K)

F_p^+ = hole flux ($\text{cm}^{-2}\text{s}^{-1}$) ↩

g_p = hole generation rate ($\text{cm}^{-3}\text{s}^{-1}$) ↩

τ_{pt} = hole lifetime, thermal-equilibrium and excess together (s) ↩

δn = excess electron concentration, equal to δp (cm^{-3}) ↩

g = generation rate ($\text{cm}^{-3}\text{s}^{-1}$) ↩

R = recombination rate ($\text{cm}^{-3}\text{s}^{-1}$) ↩

δp = excess hole concentration (cm^{-3}) ↩

E_0 = applied electric field (V/cm) ↩

E_{Fn} = quasi-Fermi level for electrons (eV) ↩

E_{Fp} = quasi-Fermi level for holes (eV) ↩

α_r = band-to-band recombination constant (cm^3/s) ↩

C_n = electron capture constant (cm^3/s) ↩

C_p = hole capture constant (cm^3/s) ↩

N_t = trap density (cm^{-3}) ↩

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n' = electron concentration if E_F were at the trap level E_t (cm^{-3}) ↩

p' = hole concentration if E_F were at E_t (cm^{-3}) ↩

J_n = electron diffusion current density (A/cm^2) ↩

J_p = hole diffusion current density (A/cm^2) ↩

ϕ = electric potential (V) ↩

$\rho(x)$ = charge density (C/cm^3) ↩

ϵ_s = permittivity of the semiconductor, $11.7\epsilon_0$ for Si (F/cm) ↩

N_0 = pairs created per unit area at $x = 0$, $t = 0$ (cm^{-2}) ↩

n = total electron concentration (cm^{-3}) ↩

p = total hole concentration (cm^{-3}) ↩

Questions:

No.	Paper	Marks	Type	Question
Steady state				
1	20-21 3(d)	10	numerical	For the following questions: (i) Solve the ambipolar transport equation for a semiconductor with $N_a = 5 \times 10^{16}$ cm^{-3} for $x > 0$ at steady state with zero electric field where excess charges are created only at $x = 0$. ($\delta n_0 = 5 \times 10^{15}$ cm^{-3}). (ii) Find the distance where the excess charge will drop to 10×10^{13} cm^{-3}. (Assume $\tau_{n0} = 7 \times 10^{-7}$ s, $\tau_{p0} = 5 \times 10^{-7}$ s, $D_n = 30$ cm^2 s^{-1}, $D_p = 25$ cm^2 s^{-1}) (iii) Plot n and p as function of x in log scale.
2	19-20 5(a)	7	derivation	Consider a p-type semiconductor that is homogeneous and infinite in extent. Assume a zero applied electric field. For a one-dimensional crystal, assume that excess carriers are being generated at $x = 0$ only. The excess carriers being generated at $x = 0$ will being *[sic: should be will begin]* diffusing in both the $-x$ and $+x$ directions. Calculate the steady-state excess carrier concentration as a function of x.
3	19-20 TT2 8(a)	6	derivation repeats 19-20 5(a)	Consider a p-type semiconductor that is homogeneous and infinite in extent. Assume a zero applied electric field. For a one-dimensional crystal, assume that excess carriers are being generated at $x = 0$ only. The excess carriers being generated at $x = 0$ will begin diffusing in both the $-x$ and $+x$ directions. Calculate the steady-state excess carrier concentration as a function of x.
4	18-19 5(c)	6	derivation	Show that diffusion length is the average length a carrier diffuses before recombination.
Dielectric relaxation				
5	19-20 3(c)	7	derivation	Explain dielectric relaxation time. Find the equation for dielectric relaxation.
Ambipolar transport				
6	19-20 4(b)	7	derivation	Define ambipolar transport. Find ambipolar transport equation. Hence show that diffuse and drift behavior of excess majority carrier is determined by excess minority carrier.
7	19-20 TT2 8(b)	7	derivation repeats 19-20 4(b)	Define ambipolar transport. Find ambipolar transport equation. Hence show that diffuse and drift behavior of excess majority carrier is determined by excess minority carrier.
Time and space dependence				
8	19-20 4(d)	6	theory	With proper graphs, explain time and spatial dependence of excess carrier concentration.
Quasi-Fermi levels				
9	19-20 5(c)	4	numerical	Consider an n-type semiconductor at $T = 300$ K with carrier concentrations of $n_o = 10^{15}$ cm^{-3}, $n_i = 10^{10}$ cm^{-3} and $p_o = 10^5$ cm^{-3}. In non-equilibrium, assume that the excess carrier concentrations are $\delta n = \delta p = 10^{-13}$ *[sic: should be* 10^{13}*]* cm^{-3}. Calculate the quasi-Fermi energy levels.
Generation and recombination				
10	19-20 TT2 8(c)	7	theory	Briefly describe different types of generation and recombination processes. Use appropriate diagram if necessary.
Minority carriers				
11	20-21 1(b)	5	theory	How do you define a minority carrier? Why does it diffuse? What is Poisson's equation and why is it important in PN junction?

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No.	Paper	Marks	Type	Question
Carrier pulse				
12	20-21 5(a)	5	theory	Write the ambipolar equation for $g' = 0$ for n type semiconductor with constant electric field E_0. At time $t = 0$, a number of electron-hole pair is generated at $x = 0$. Sketch excess electron concentration as a function of distance at different time.
Excess carriers				
13	20-21 TT1 7(c)	1	theory	What are n, p, n_o and p_o?

4.6. PN junction:

72 marks, 12 questions, in 3 of 4 papers. Book: Neamen Ch. 7, 8.

Most often numerical (5 of 12). Subtopics: Switching transient; Junction capacitance; Ideal diode current; Depletion width; Built-in potential; Band diagram; Biased junction.

Formulas:

Subtopic	Formula	Equation	Used in
Switching transient (7)	**Storage time**	$\text{erf}\sqrt{\frac{t_s}{\tau_{p0}}} = \frac{I_F}{I_F + I_R}$	18-19 2(a)
Junction capacitance (6)	**Built-in potential**	$V_{bi} = \frac{kT}{e} \ln\left(\frac{N_a N_d}{n_i^2}\right)$	19-20 3(b), 19-20 4(c), 19-20 5(d), 18-19 2(d), 18-19 8(a)
	Total width	$W = \left\{\frac{2\epsilon_s(V_{bi} + V_R)}{e}\left[\frac{N_a + N_d}{N_a N_d}\right]\right\}^{1/2}$	19-20 4(c), 19-20 5(d), 18-19 2(d), 18-19 8(a)
	Junction capacitance	$C' = \frac{\epsilon_s}{W} = \left\{\frac{e\epsilon_s N_a N_d}{2(V_{bi} + V_R)(N_a + N_d)}\right\}^{1/2}$	18-19 2(d)
Ideal diode current (15)	**Ideal diode**	$J = \left[\frac{eD_p p_{n0}}{L_p} + \frac{eD_n n_{p0}}{L_n}\right]\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$	19-20 5(b), 18-19 4(a)
	Law of the junction	$p_n(x_n) = p_{n0} e^{eV_a/kT}$, $n_p(-x_p) = n_{p0} e^{eV_a/kT}$	19-20 5(b)
	Current components	$J_p(x) = J_p(x_n) e^{(x_n - x)/L_p}$, $J_n(x) = J - J_p(x)$	19-20 5(b)
Depletion width (25)	**Charge balance**	$N_a x_p = N_d x_n$	20-21 2(c), 18-19 8(a)
	Space charge widths	$x_n = \left[\frac{2\epsilon_s V_{bi}}{e}\frac{N_a}{N_d}\frac{1}{N_a + N_d}\right]^{1/2}$, $x_p = \left[\frac{2\epsilon_s V_{bi}}{e}\frac{N_d}{N_a}\frac{1}{N_a + N_d}\right]^{1/2}$	18-19 8(a)
	n_i at another temperature	$n_i(T) = n_i(300)\left(\frac{T}{300}\right)^{3/2} \exp\left[-\frac{E_g}{2k}\left(\frac{1}{T} - \frac{1}{300}\right)\right]$	19-20 4(c)
	Split of W	$x_n = W\frac{N_a}{N_a + N_d}$, $x_p = W\frac{N_d}{N_a + N_d}$	20-21 2(c), 19-20 4(c), 19-20 5(d)
	Peak field	$\lvert E_{max}\rvert = \frac{eN_d x_n}{\epsilon_s} = \frac{2(V_{bi} + V_R)}{W}$	19-20 4(c), 19-20 5(d)
	Depletion width	$W = x_n + x_p = \left\{\frac{2\epsilon_s V_{bi}}{e}\left[\frac{N_a + N_d}{N_a N_d}\right]\right\}^{1/2}$	20-21 2(c)

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Subtopic	Formula	Equation	Used in
	Peak field	$E_{max} = \frac{eN_d x_n}{\epsilon_s} = \frac{eN_a x_p}{\epsilon_s} = \frac{2V_{bi}}{W}$	20-21 6(d)
Built-in potential (4)	Current density	$J = en\mu_n E + ep\mu_p E + eD_n \frac{dn}{dx} - eD_p \frac{dp}{dx}$	19-20 3(b)
	Barrier and minority carriers	$n_{p0} = n_{n0} \exp\left(-\frac{eV_{bi}}{kT}\right)$	19-20 3(b), 19-20 5(b)
Band diagram (5)	**Fermi potentials**	$\phi_{Fp} = V_t \ln \frac{N_a}{n_i}$, $\lvert\phi_{Fn}\rvert = V_t \ln \frac{N_d}{n_i}$	20-21 3(b)
	Built-in potential	$V_{bi} = \lvert\phi_{Fn}\rvert + \lvert\phi_{Fp}\rvert = V_t \ln \frac{N_a N_d}{n_i^2}$	20-21 3(b), 20-21 6(d)
Biased junction (10)	**Law of the junction**	$p_n(x_n) = p_{n0} \exp\left(\frac{eV_a}{kT}\right)$, $n_p(-x_p) = n_{p0} \exp\left(\frac{eV_a}{kT}\right)$	20-21 4(d)
	Majority carriers at the edge	$n_n = N_d + \delta p_n$	20-21 4(d)

Symbols:

t_s = storage time (s) ↩

I_F = forward current before switching (A) ↩

I_R = reverse current while the stored charge is removed (A) ↩

Q_p = stored minority-carrier charge (C)

V_{bi} = built-in potential barrier (V) ↩

W = total width of the space-charge (depletion) region (cm); total depletion width (cm) ↩

V_R = reverse-bias voltage (V) ↩

ϵ_s = permittivity of the semiconductor, $11.7\epsilon_0$ for Si (F/cm) ↩

C' = junction capacitance per unit area (F/cm^2) ↩

p_{n0} = thermal-equilibrium hole (minority) concentration in the n region, n_i^2/N_d (cm^{-3}) ↩

n_{p0} = thermal-equilibrium electron (minority) concentration in the p region (cm^{-3}) ↩

L_n = electron diffusion length (cm) ↩

V_a = applied forward voltage (V) ↩

L_p = hole diffusion length, $\sqrt{D_p \tau_{p0}}$ (cm) ↩

$p_n(x_n)$ = hole concentration at the n-side edge of the depletion region (cm^{-3}) ↩

$n_p(-x_p)$ = electron concentration at the p-side edge of the depletion region (cm^{-3}) ↩

$J_p(x)$ = hole (minority) diffusion current density in the n region (A/cm^2) ↩

$J_n(x)$ = electron current density (A/cm^2) ↩

x_n = space-charge width on the n side (cm); depletion width on the n side (cm) ↩

x_p = space-charge width on the p side (cm); depletion width on the p side (cm) ↩

$n_i(300)$ = intrinsic concentration at 300 K (cm^{-3}) ↩

E_{max} = peak electric field, at the metallurgical junction (V/cm) ↩

D_n = electron diffusion coefficient (cm^2/s) ↩

D_p = hole diffusion coefficient (cm^2/s) ↩

n_{n0} = thermal-equilibrium electron (majority) concentration in the n region (cm^{-3}) ↩

ϕ_{Fp} = Fermi potential of the p side, $(E_{Fi} - E_F)/e$ (V) ↩

ϕ_{Fn} = Fermi potential of the n side, $\lvert\phi_{Fn}\rvert = (E_F - E_{Fi})/e$ (V) ↩

V_t = thermal voltage, kT/e (V) ↩

n_n = electron (majority) concentration at the n-side edge (cm^{-3}) ↩

δp_n = excess holes injected there (cm^{-3}) ↩

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Questions:

No.	Paper	Marks	Type	Question
Switching transient				
1	18-19 2(a)	7	theory	What is storage delay time? What is the impact of reverse recovery time of a junction in switching application?
Junction capacitance				
2	18-19 2(d)	6	theory	Wat *[sic: should be "What"]* is the depletion capacitance of a PN junction? Explain its variation with reverse bias voltage.
Ideal diode current				
3	19-20 5(b)	8	derivation repeats 18-19 4(a)	Find ideal pn junction current and sketch ideal electron and hole current components.
4	18-19 4(a)	7	diagram	Sketch the ideal electron and hole current components through a *pn* junction under forward bias.
Depletion width				
5	20-21 2(c)	0	derivation	Using the expression for V_{bi}, find the width of the depletion region.
6	20-21 6(d)	10	numerical repeats 18-19 8(a)	A silicon PN junction at $T = 300$ K with zero applied bias has doping concentration of $N_d = 5 \times 10^{16}$ cm^{-3} and $N_a = 5 \times 10^{15}$ cm^{-3}. Determine built-in potential, space charge width and electric field at metallurgical region ($\epsilon_t = 11.7$ *[sic: should be* $\epsilon_r = 11.7$*]*)
7	19-20 4(c)	6	numerical	For a pn junction with $N_a = 1.5 \times 10^{18}$cm^{-3}, $N_d = 1.5 \times 10^{16}$cm^{-3} and $n_i = 2 \times 10^{10}$cm^{-3} (i) calculate the built-in potential barrier V_{bi} (ii) calculate the depletion region with *[sic: should be width]* and E_{max} (Assume $\epsilon_s = 11.7\epsilon_o$) and (iii) what will be the width at liquid nitrogen temperature (80K).
8	19-20 5(d)	6	numerical	An abrupt silicon pn junction has dopant concentrations of $N_a = 2 \times 10^{16}$ cm^{-3} and $N_d = 2 \times 10^{15}$ cm^{-3} at $T = 300$ K. Calculate: (i) V_{bi} (ii) W at $V_R = 0$ and $V_R = 8$ V and (iii) the maximum electric field in the space charge region at $V_R = 0$ and $V_R = 8$ V
9	18-19 TT2 8(a)	3	numerical	A silicon pn junction at T = 300 K with zero applied bias has doping concentrations of $N_d = 5 \times 10^{16}$ cm^{-3} and $N_a = 5 \times 10^{20}$ *[sic: should be* 5×10^{15}*]* cm^{-3}. Determine x_n, x_p and W.
Built-in potential				
10	19-20 3(b)	4	theory	Explain how barrier potential maintains equilibrium p, n regions.
Band diagram				
11	20-21 3(b)	5	theory	Sketch Energy-band diagram of a PN junction in thermal equilibrium and show P region, N region, E_C, E_V, E_F, E_{Fi}, $e\phi_{Fn}$, $e\phi_{Fp}$ and eV_{bi}.
Biased junction				
12	20-21 4(d)	10	numerical	Consider a PN junction at room temperature with $n_i = 1.5 \times 10^{10}$ cm^{-3}, assume doping in N side is 1×10^6 cm^{-3} *[sic: should be* 1×10^{16} *cm*$^{-3}$*]*. What will be the minority carrier hole concentrations at the edge in N side if forward bias and reverse bias of 0.7 V is applied? Do the same for majority carrier electron concentration in N side.

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4.7. Tunnel diode:

11 marks, 2 questions, in 2 of 4 papers. Book: Neamen Ch. 8, 15.

Most often theory (2 of 2). Subtopics: Operation.

Questions:

No.	Paper	Marks	Type	Question
Operation				
1	20-21 5(c)	5	theory repeats 18-19 4(e)	Explain the working principle of Tunnel diode. Draw its characteristics curve.
2	18-19 4(e)	6	theory	Explain the working of Tunnel diode. Draw its characteristics curve.

4.8. Metal-semiconductor contacts:

16 marks, 3 questions, in 2 of 4 papers. Book: Neamen Ch. 9.

Most often theory (3 of 3). Subtopics: Schottky and ohmic contacts.

Formulas:

Subtopic	Formula	Equation	Used in
Schottky and ohmic contacts (16)	**Schottky barrier**	$\phi_{B0} = \phi_m - \chi,\ V_{bi} = \phi_{B0} - \phi_n,\ \phi_n = \frac{kT}{e}\ln\frac{N_c}{N_d}$	18-19 6(d)
	Thermionic emission	$J = A^*T^2\exp\left(-\frac{e\phi_{B0}}{kT}\right)\left[\exp\left(\frac{eV_a}{kT}\right)-1\right]$	18-19 6(d), 18-19 4(c)
	Schottky barrier	$\phi_{B0} = \phi_m - \chi,\ V_{bi} = \phi_{B0} - \phi_n$	19-20 1(d)
	Thermionic emission	$J = J_{sT}\left[\exp\left(\frac{eV_a}{kT}\right)-1\right],\ J_{sT} = A^*T^2\exp\left(-\frac{e\phi_{Bn}}{kT}\right)$	19-20 1(d)

Symbols:

ϕ_{B0} = ideal Schottky barrier height (V) ↩

ϕ_m = metal work function (V) ↩

χ = electron affinity (V), 4.01 V for Si ↩

ϕ_n = depth of the Fermi level below E_c, $(E_c - E_F)/e$ (V) ↩

A^* = effective Richardson constant ($\mathrm{A/cm^2K^2}$) ↩

V_{bi} = built-in potential barrier (V) ↩

J = current density ($\mathrm{A/cm^2}$) ↩

J_{sT} = reverse saturation current density ($\mathrm{A/cm^2}$) ↩

e = magnitude of the electron charge (1.6×10^{-19} C) ↩

V_a = applied forward voltage (V) ↩

ϕ_{Bn} = Schottky barrier height (V) ↩

Questions:

No.	Paper	Marks	Type	Question
Schottky and ohmic contacts				
1	19-20 1(d)	6	theory repeats 18-19 4(c)	Explain Schottky barrier. How it differs from ohmic contact? Explain.
2	18-19 4(c)	4	theory	Differentiate between ohmic and rectifying contacts.
3	18-19 6(d)	6	theory	With suitable energy band diagrams, explain Schottky contact.

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4.9. MOS structure:

54 marks, 9 questions, in 3 of 4 papers. Book: Neamen Ch. 10.

Most often theory (6 of 9). Subtopics: Band diagrams; Threshold voltage; Body effect; C-V characteristics; MOSFET advantages; Double gate; Inversion.

Formulas:

Subtopic	Formula	Equation	Used in
Band diagrams (13)	**Fermi potential**	$\phi_{fp} = V_t \ln \frac{N_a}{n_i}$	20-21 4(b), 19-20 6(a), 18-19 3(a), 18-19 3(b)
Threshold voltage (18)	**Threshold voltage**	$V_{TN} = \frac{\lvert Q'_{SD}(\max)\rvert - Q'_{ss}}{C_{ox}} + \phi_{ms} + 2\phi_{fp}$, $x_{dT} = \left(\frac{4\epsilon_s \phi_{fp}}{eN_a}\right)^{1/2}$	18-19 3(b)
	Maximum depletion	$x_{dT} = \left(\frac{4\epsilon_s \phi_{fp}}{eN_a}\right)^{1/2}$, $\lvert Q'_{SD}(max)\rvert = eN_a x_{dT}$	19-20 6(a)
	Oxide field limit	$\epsilon_{ox} E_{ox} = \lvert Q'_{SD}\rvert + \lvert Q'_I\rvert$	19-20 6(a)
	Threshold voltage	$V_T = V_{FB} + 2\phi_{fp} + \frac{\lvert Q'_{SD}(max)\rvert}{C_{ox}}$, $C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$	19-20 6(a)
	Gate voltage above threshold	$V_G = V_T + \frac{\lvert Q'_I\rvert}{C_{ox}} = V_{FB} + 2\phi_{fp} + E_{ox} t_{ox}$	19-20 6(a)
Body effect (5)	**Body effect**	$\Delta V_T = \gamma \left[\sqrt{2\phi_{fp} + V_{SB}} - \sqrt{2\phi_{fp}}\right]$, $\gamma = \frac{\sqrt{2e\epsilon_s N_a}}{C_{ox}}$	18-19 3(c)
C-V characteristics (5)	**MOS capacitance**	$\frac{1}{C'} = \frac{1}{C_{ox}} + \frac{1}{C'_{SD}}$, $C'_{min} = \frac{\epsilon_{ox}}{t_{ox} + (\epsilon_{ox}/\epsilon_s) x_{dT}}$	18-19 5(b)
Inversion (5)	**Threshold inversion**	$\phi_s = 2\phi_{fp}$	20-21 6(c)
	Maximum depletion width	$x_{dT} = \left[\frac{4\epsilon_s \phi_{fp}}{eN_a}\right]^{1/2}$	20-21 6(c)

Symbols:

ϕ_{fp} = Fermi potential of the p-type substrate, $(E_{Fi} - E_F)/e$ (V) ↩

V_t = thermal voltage, kT/e (0.0259 V at 300 K) ↩

ϕ_s = surface potential, the band bending (V) ↩

V_{TN} = threshold voltage of the n-channel MOSFET (V) ↩

$Q'_{SD}(\max)$ = maximum space-charge (depletion) charge per unit area (C/cm^2) ↩

Q'_{ss} = fixed oxide charge per unit area (C/cm^2) ↩

C_{ox} = oxide capacitance per unit area, ϵ_{ox}/t_{ox} (F/cm^2) ↩

ϕ_{ms} = metal-semiconductor work function difference (V) ↩

x_{dT} = maximum space-charge width under the gate (cm) ↩

$Q'_{SD}(max)$ = maximum space-charge (depletion) charge per unit area (C/cm^2) ↩

ϵ_{ox} = oxide permittivity, $3.9\epsilon_0$ (F/cm) ↩

E_{ox} = electric field in the oxide (V/cm) ↩

Q'_I = inversion charge per unit area (C/cm^2) ↩

V_T = threshold voltage (V) ↩

V_{FB} = flat-band voltage (V) ↩

t_{ox} = oxide thickness (cm) ↩

V_G = gate voltage (V) ↩

ΔV_T = shift of the threshold voltage (V) ↩

γ = body-effect coefficient ($V^{1/2}$) ↩

V_{SB} = source-to-body reverse bias (V) ↩

C'_{SD} = space-charge capacitance per unit area (F/cm^2) ↩

C'_{min} = minimum capacitance per unit area (F/cm^2) ↩

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Questions:

No.	Paper	Marks	Type	Question
Band diagrams				
1	20-21 4(b)	5	theory repeats 18-19 3(a)	Draw the energy band diagram of a Metal-Oxide-Semiconductor structure for the three modes of operation of MOSFET and thus explain how the channel is formed.
2	18-19 3(a)	8	diagram	With brief description sketch the energy band diagram of a MOS structure at different bias voltage. (Use no bias condition, positive bias voltage and negative bias voltage).
Threshold voltage				
3	19-20 6(a)	12	numerical	The body of a nanoscale N-channel MOSFET is doped at $N_A = 5^{18}$ *[sic: should be* 5×10^{18}*]* cm^{-3}. (i) Determine the maximum inversion-layer charge per unit area (Q_I) that can be reached so that the electric field in the oxide remains below $E_{ox} = 0.6$ V/nm. Neglect any interface-trap and gate-oxide charge. (ii) If the channel length and the channel width are $L = 50$ nm and $W = 500$ nm, respectively, determine how many electrons are creating this inversion layer. (iii) Calculate the threshold voltage of a MOSFET having gate-oxide thickness $t_{ox} = 1$ nm and flat-band voltage $V_{FB} = 1.0$ V. (iv) What gate voltage is needed to form the maximum inversion-layer charge?
4	18-19 3(b)	6	theory	What are the parameters that affect the magnitude of threshold voltage in a MOSFET?
Body effect				
5	18-19 3(c)	5	theory	What is body effect of MOSFET?
C-V characteristics				
6	18-19 5(b)	5	diagram	Draw the ideal low-frequency capacitance versus gate voltage curve of a MOS capacitor with p-type substrate and indicate different regions.
MOSFET advantages				
7	19-20 6(c)	3	theory	Mention some advantages of using MOSFET.
Double gate				
8	20-21 6(a)	5	theory	Draw the schematic of a double-gate MOS structure. What are the benefits of such structure over single-gate MOS?
Inversion				
9	20-21 6(c)	5	theory	Explain production of inverse layer at oxide and semiconductor layer with proper sketches of energy band diagram.

4.10. Bipolar transistor:

47 marks, 9 questions, in 3 of 4 papers. Book: Neamen Ch. 12.

Most often theory (7 of 9). Subtopics: Amplification; Base current; Base width modulation; Emitter efficiency; Current crowding; Operating modes; Minority distribution.

Formulas:

Subtopic	Formula	Equation	Used in
Amplification (6)	**Collector current**	$i_C = I_S \exp\left(\frac{v_{BE}}{V_t}\right),\ A_v = -g_m R_C = -\frac{I_C R_C}{V_t}$	18-19 3(d)
Base current (3)	Base current	$I_B \approx I_{pE} + I_{RB} + I_R - I_{CBO}$	18-19 4(b)
Base width modulation (5)	**Early effect**	$\frac{dI_C}{dV_{CE}} = \frac{I_C}{V_{CE} + V_A},\ J_C \approx \frac{eD_B n_{B0}}{x_B} e^{eV_{BE}/kT}$	18-19 4(d)
Emitter efficiency (10)	**Injection efficiency**	$\gamma \approx \frac{1}{1 + \frac{N_B}{N_E}\frac{D_E}{D_B}\frac{x_B}{x_E}}$	18-19 6(b)

Subtopic	Formula	Equation	Used in
	Emitter injection efficiency	$\gamma = \dfrac{J_{nE}}{J_{nE}+J_{pE}} \approx \dfrac{1}{1+\dfrac{N_B}{N_E}\dfrac{D_E}{D_B}\dfrac{x_B}{x_E}}$	19-20 6(d)
Current crowding (7)	Current crowding	$\Delta V = \dfrac{I_B}{2}R$, injection $\propto e^{eV_{BE}/kT}$	18-19 6(c)
Operating modes (11)	**BJT modes**	active: $V_{BE}>0$, $V_{BC}<0$; saturation: both >0; cutoff: both <0; inverse: $V_{BE}<0$, $V_{BC}>0$	19-20 6(b)
	Equilibrium minority carriers	$p_{E0} = \dfrac{n_i^2}{N_E}$, $n_{B0} = \dfrac{n_i^2}{N_B}$, $p_{C0} = \dfrac{n_i^2}{N_C}$	20-21 4(c)
	Four modes	active: $V_{BE}>0$, $V_{CB}>0$; saturation: $V_{BE}>0$, $V_{CB}<0$; cutoff: $V_{BE}<0$, $V_{CB}>0$; inverse: $V_{BE}<0$, $V_{CB}<0$	20-21 4(c)
Minority distribution (5)	**Base edges**	$n_B(0) = n_{B0}\exp\left(\dfrac{V_{BE}}{V_t}\right)$, $n_B(x_B) = n_{B0}\exp\left(\dfrac{V_{BC}}{V_t}\right)$	20-21 5(b)
	Emitter and collector edges	$p_E(0) = p_{E0}\exp\left(\dfrac{V_{BE}}{V_t}\right)$, $p_C(0) = p_{C0}\exp\left(\dfrac{V_{BC}}{V_t}\right)$	20-21 5(b)

Symbols:

i_C = collector current (A) ↩

I_S = saturation current (A) ↩

v_{BE} = base-emitter voltage (V) ↩

A_v = voltage gain ↩

g_m = transconductance, I_C/V_t (A/V) ↩

R_C = collector load resistance (Ω) ↩

I_C = dc collector current (A) ↩

I_B = base current (A) ↩

I_{pE} = hole current injected from the base into the emitter (A) ↩

I_{RB} = recombination current in the base (A) ↩

I_R = recombination current in the B-E space-charge region (A) ↩

I_{CBO} = B-C reverse-bias current (A) ↩

V_{CE} = collector-emitter voltage (V) ↩

V_A = Early voltage (V) ↩

J_C = collector current density (A/cm^2) ↩

D_B = minority-carrier diffusion coefficient in the base (cm^2/s) ↩

n_{B0} = thermal-equilibrium minority-carrier concentration in the base (cm^{-3}); equilibrium electron concentration in the base (cm^{-3}) ↩

x_B = neutral base width (cm) ↩

V_{BE} = base-emitter voltage (V) ↩

γ = emitter injection efficiency ↩

N_B = base doping (cm^{-3}) ↩

N_E = emitter doping (cm^{-3}) ↩

D_E = minority-carrier diffusion coefficient in the emitter (cm^2/s) ↩

x_E = neutral emitter width (cm) ↩

J_{nE} = electron current density injected from the emitter into the base (A/cm^2) ↩

J_{pE} = hole current density injected from the base into the emitter (A/cm^2) ↩

ΔV = drop of V_{BE} from the emitter edge to its centre (V) ↩

R = lateral base resistance under half the emitter (Ω) ↩

S = emitter stripe width (cm)

V_{BC} = base-collector voltage (V) ↩

β = common-emitter current gain

p_{E0} = equilibrium hole concentration in the emitter (cm^{-3}) ↩

p_{C0} = equilibrium hole concentration in the collector (cm^{-3}) ↩

N_C = collector doping (cm^{-3}) ↩

V_{CB} = collector-base voltage (V) ↩

$n_B(0)$ = electron concentration at the emitter edge of the base (cm^{-3}) ↩

$n_B(x_B)$ = electron concentration at the collector edge of the base (cm^{-3}) ↩

$p_E(0)$ = hole concentration at the edge of the emitter (cm^{-3}) ↩

$p_C(0)$ = hole concentration at the edge of the collector (cm^{-3}) ↩

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Questions:

No.	Paper	Marks	Type	Question
Amplification				
1	18-19 3(d)	6	theory	Explain the amplification action of a transistor.
Base current				
2	18-19 4(b)	3	theory	What are the factors which cause base current in a transistor?
Base width modulation				
3	18-19 4(d)	5	theory	What is base width modulation?
Emitter efficiency				
4	19-20 6(d)	4	theory	Does the doping level in the base of a BJT affect the emitter efficiency γ_E? If so, why?
5	18-19 6(b)	6	numerical	Design the ratio of emitter doping to base doping in order to achieve an emitter injection efficiency factor of $\gamma = 0.9967$. Assume a *npn* bipolar transistor, and also consider $D_E = D_B$, $L_E = L_B$ and $x_E = X_B$.
Current crowding				
6	18-19 6(c)	7	theory	What is current crowding? What is the impact of current crowding in high power devices?
Operating modes				
7	20-21 4(c)	5	theory	Show the four operating modes of a bipolar transistor as a function of V_{BE} and V_{CB}. Draw the minority carrier distribution of the four modes of a NPN transistor with proper labelling of all the quantities.
8	19-20 6(b)	6	numerical	Determine the mode of operation of an NPN BJT with $\beta \approx 450$, if it is known that: (i) $V_{BE} = 0.7V$, $V_{CE} = 5.2V$ (ii) $V_{BE} = 0.7V$, $V_{CE} = 0.2V$ (iii) $V_{BE} = 0.8V$, $V_{BC} = 0.3V$ (iv) $V_{BE} = -0.8V$, $V_{BC} = 0.7V$ (v) $I_C = 455mA$, $I_B = 1mA$ (vi) $I_C = 455mA$, $I_E = 502mA$
Minority distribution				
9	20-21 5(b)	5	theory	For NPN transistor, plot minority carrier concentration in base, emitter, and collector as a function of space. What are the minority carrier concentrations at the boundaries of base, emitter, and collector?

4.11. JFET:

13 marks, 2 questions, in 1 of 4 papers. Book: Neamen Ch. 13.

Most often numerical (2 of 2). Subtopics: Pinch-off; Frequency response.

Formulas:

Subtopic	Formula	Equation	Used in
Pinch-off (7)	**Pinch-off voltage**	$V_{p0} = \dfrac{ea^2 N_d}{2\epsilon_s}$, $V_p = V_{bi} - V_{p0}$	18-19 5(d)
Frequency response (6)	**Cutoff frequency**	$f_T = \dfrac{g_m}{2\pi C_G} = \dfrac{e\mu_n N_d a^2}{2\pi\epsilon_s L^2}$	18-19 6(a)

Symbols:

V_{p0} = internal pinch-off voltage (V) ↩

a = metallurgical channel thickness (cm)

C_G = gate capacitance, $C_{gs} + C_{gd}$ (F) ↩

V_p = pinch-off voltage, gate to source (V) ↩

f_T = cutoff frequency (Hz) ↩

L = channel length (cm)

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Questions:

No.	Paper	Marks	Type	Question
Pinch-off				
1	18-19 5(d)	7	numerical	A silicon n-channel JFET at $T = 300K$ has a gate doping concentration of $N_a = 10^{18} cm^{-3}$ and a channel doping concentration of $N_d = 2 \times 10^{16} cm^{-3}$. Determine the metallurgical channel thickness, a, such that the pinchoff voltage is $V_p = -2.50\ V$.
Frequency response				
2	18-19 6(a)	6	numerical	The parameters of an n-channel silicon JFET are $\mu_n = 1000\ cm^2/V - s$, $N_d = 5 \times 10^{15} cm^{-3}$, $a = 0.50\ \mu m$, and $L = 2\ \mu m$. Determine the cutoff frequency.

5. Theory points:

What a full answer covers, for every theory question, in book order. The link opens the full answer.

No.	Paper	Marks	Subtopic	Points to cover
Crystal structure				
1	19-20 1(a)	6	Classes of solids	Amorphous: order only within a few atomic distances, no grains (glass, a-Si). Polycrystalline: high order inside grains of different orientation, separated by grain boundaries (polysilicon gates). Single crystal: one lattice through the whole volume, no grain boundaries, best electrical properties (Si wafers).
Quantum basics				
2	21-22 7(a)	3	Fermi probability	$f_F(E) = 1/[1 + e^{(E-E_F)/kT}]$ is the probability that a state at E holds an electron (Pauli: one per state, particles not told apart). Step at 0 K; 1/2 at E_F for any T, rounded over a few kT; $1 - f_F$ is the hole probability, symmetric about E_F; Boltzmann form $e^{-(E-E_F)/kT}$ for $E - E_F > 3kT$.
3	20-21 TT2 8(a)	1	Fermi probability	$f_F(E) = 1/[1 + \exp((E - E_F)/kT)]$: probability that a state at E is occupied; 1/2 at E_F, a step at 0 K; $1 - f_F$ is the probability of a hole.
4	18-19 1(c)	5	Fermi probability	At 0 K f_F is a step: every state below E_F full, every state above empty. In a semiconductor E_F lies in the gap, so the valence band is completely full and the conduction band empty: the highest electron energy is E_v, the top of the valence band (in a metal it would be E_F).
5	18-19 2(b)	6	Fermi probability	Fermi energy E_F: the energy at which a state is occupied with probability 1/2; at 0 K all states below it are full; its position in the gap sets n_0 and p_0. kT at 300 K: $(1.38 \times 10^{-23})(300)/(1.6 \times 10^{-19}) = 0.0259$ eV $\approx 1/40$ eV.
6	18-19 7(a)	2	Density of states	Density of states $g(E)$: allowed states per unit energy per unit volume, $g_c = 4\pi(2m_n^*)^{3/2}\sqrt{E - E_c}/h^3$. Carrier concentration: free carriers per unit volume, $n_0 = \int g_c f_F\, dE = N_c e^{-(E_c - E_F)/kT}$.
7	18-19 7(c)	3	Effective mass	Effective mass: the mass that makes Newton's law work for an electron in the crystal, including the lattice forces: $1/m^* = (1/\hbar^2) d^2E/dk^2$. Band B is flatter (smaller curvature), so it gives the heavier effective mass (about 2.4 times band A from the figure).
8	18-19 7(d)	3	Energy bands	Isolated atoms have discrete levels; brought together, wave functions overlap and Pauli forces each level to split into N levels, forming bands. Bloch's theorem $\psi = u(x)e^{jkx}$ in the periodic Kronig-Penney potential gives $P' \sin \alpha a/\alpha a + \cos \alpha a = \cos ka$: allowed where the left side lies within ± 1, so allowed bands separated by forbidden gaps.

No.	Paper	Marks	Subtopic	Points to cover
Semiconductor in equilibrium				
9	21-22 7(b)	0	Ionization	Energy to free a donor's fifth electron ($E_c - E_d$) or to fill an acceptor's missing bond ($E_a - E_v$), leaving a fixed ion and a free carrier. Hydrogen model: $E_{ion} = 13.6(m^*/m_0)/\epsilon_r^2 = 25.8$ meV in Si; measured 0.045 eV (P, B), 0.05 eV (As); close to kT, so dopants are fully ionized at 300 K.
10	20-21 1(a)	5	Fermi level	$E_F - E_{Fi} = kT\ln(N_d/n_i)$ in n-type, $E_{Fi} - E_F = kT\ln(N_a/n_i)$ in p-type; on a log doping axis both are straight lines of 0.06 eV per decade (0.23, 0.35, 0.47, 0.53 eV at 10^{14} to 10^{19} cm^{-3}); E_F moves towards E_c with donors and E_v with acceptors, reaching the band edge at degeneracy.
11	20-21 2(a)	5	Compensation	Both donors and acceptors in one region; donor electrons fall into the acceptor states, so donors are ionized (+) and acceptors filled (−). For $N_d > N_a$: n-type, $n_0 \approx N_d - N_a$, E_F above E_{Fi}; for $N_a > N_d$: p-type; for $N_a = N_d$: behaves intrinsic.
12	20-21 TT1 7(a)	1	Ionization	Essentially all (complete ionization) because $E_c - E_d = 0.045$ eV $< 2kT$ and $N_c \gg N_d$; fraction still on donors $1/[1 + (N_c/2N_d)e^{-(E_c-E_d)/kT}] = 0.40$ % for P at 10^{16} cm^{-3} (99.6 % ionized); B acceptors at 10^{16} cm^{-3} about 97.9 % ionized ($g = 4$).
13	20-21 TT1 7(b)	1	Ionization	$f_d = 1/[1 + \frac{1}{2}\exp((E_d - E_F)/kT)]$, $n_d = N_d - N_d^+$; the $\frac{1}{2}$ ($1/g$, $g = 2$) comes from the two spin states of a donor level that holds only one electron; $f_d = 2/3$ at $E_d = E_F$.
14	20-21 TT1 7(d)	1	Dopants	Yes, in a III-V compound such as GaAs: Si (4 valence electrons) on a Ga site (3) has one left over and is a donor (0.0058 eV); on an As site (5) it is an acceptor, so Si is amphoteric, mostly a donor in GaAs; in Si it is the host atom.
15	20-21 TT1 7(e)	1	Carrier concentrations	True: $n_i \propto e^{-E_g/2kT}$ rises steeply and exceeds the doping at high T, so $n_0 \approx p_0 \approx n_i$ and $E_F \to E_{Fi}$; for $N_d = 10^{15}$ cm^{-3}, $n_i = N_d$ at about 565 K.
16	20-21 TT2 8(b)	1	Compensation	Both donors and acceptors in the same region; donor electrons fill the acceptor states, so only the net doping gives carriers: $n_0 \approx N_d - N_a$ if $N_d > N_a$, $p_0 \approx N_a - N_d$ if $N_a > N_d$, intrinsic-like if equal.
17	20-21 TT2 8(d)	1	Dopants	Yes, in GaAs: Si on a Ga site has one valence electron more than Ga and gives it to the conduction band (0.0058 eV); on an As site it is an acceptor (amphoteric), mostly a donor; in Si it is the host.
18	20-21 TT2 8(e)	1	Degenerate	So heavily doped ($\gtrsim 10^{18}$ cm^{-3} in Si) that the impurity band overlaps the band edge and E_F lies inside the conduction band ($n_0 > N_c$) or the valence band ($p_0 > N_v$); Boltzmann and $n_0 p_0 = n_i^2$ fail; used in tunnel diodes and ohmic contacts.
19	19-20 2(c)	5	Temperature dependence	Freeze-out at low T (donors not ionized, $n_0 < N_d$), extrinsic plateau ($n_0 = N_d$, about 100 to 500 K for 10^{15} cm^{-3}), intrinsic region at high T where $n_i \propto T^{3/2}e^{-E_g/2kT}$ overtakes N_d and $n_0 \to n_i$.
20	19-20 2(d)	6	Compensation	Donor electrons drop into acceptor levels, so all dopants are ionized; $N_d > N_a$: n-type, $n_0 \approx N_d - N_a$, E_F above E_{Fi}; $N_a > N_d$: p-type, $p_0 \approx N_a - N_d$, E_F below E_{Fi}; $N_d = N_a$: intrinsic-like; mobility set by $N_d + N_a$.
21	19-20 3(a)	8	Carrier distribution	$g_c \propto \sqrt{E - E_c}$, $g_v \propto \sqrt{E_v - E}$, zero in the gap; f_F with E_F above E_{Fi} for n-type; $n(E) = g_c f_F$ and $p(E) = g_v(1 - f_F)$ peak $kT/2$ inside each band; areas $n_0 \gg p_0$.
22	19-20 TT1 7(b)	5	Fermi level	$E_F - E_{Fi} = kT\ln(n_0/n_i)$, $E_{Fi} - E_F = kT\ln(p_0/n_i)$, or $E_c - E_F = kT\ln(N_c/N_d)$, $E_F - E_v = kT\ln(N_v/N_a)$: E_F moves 0.06 eV per decade of doping towards the majority band edge, and back towards E_{Fi} as T rises and n_i grows; freeze-out at very low T.

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No.	Paper	Marks	Subtopic	Points to cover
23	18-19 1(a)	7	Temperature dependence	Neutrality $n_0 = N_d^+ + p_0$ with partial ionization. Freeze-out at low T (electrons stay on donors, $n_0 \to 0$ at 0 K); extrinsic plateau $n_0 = N_d$ (about 100 to 500 K for 10^{15} cm^{-3} in Si); intrinsic region where $n_i = \sqrt{N_c N_v} e^{-E_g/2kT}$ overtakes N_d. E_F moves from near E_c down to E_{Fi} as T rises.
24	18-19 7(b)	3	Carrier concentrations	$n_0 = \int g_c f_F \, dE$: f_F at E_c is tiny ($\sim 10^{-10}$ in Si with E_F at midgap) but there are very many states ($N_c = 2.8 \times 10^{19}$ cm^{-3}); the product gives $n_i = \sqrt{N_c N_v} e^{-E_g/2kT} \sim 10^{10}$ cm^{-3}.
Carrier transport				
25	20-21 3(c)	5	Mobility	Drift mobility $\mu = \lvert v_d \rvert / E = e\tau/m^*$. Scattering by lattice vibrations ($\mu_L \propto T^{-3/2}$) and ionized impurities ($\mu_I \propto T^{3/2}/N_I$), combined by $1/\mu = 1/\mu_L + 1/\mu_I$. The mean free time τ (about 10^{-13} s) sets the mobility and so the conductivity.
26	20-21 4(a)	5	Mobility	Lattice scattering grows with T, so $\mu_L \propto T^{-3/2}$ falls; impurity scattering weakens as carriers get faster, so $\mu_I \propto T^{3/2}/N_I$ rises; by Matthiessen's rule impurity scattering dominates at low T (mobility rises), lattice at high T (mobility falls); heavier doping lowers the curve.
27	19-20 4(a)	6	Mobility	Lattice scattering $\mu_L \propto T^{-3/2}$, impurity scattering $\mu_I \propto T^{3/2}/N_I$, $1/\mu = 1/\mu_L + 1/\mu_I$: μ rises with T at low T, falls at high T; flat at low doping (about 1350 and 480 cm^2/V s in Si), falls steeply above 10^{16} cm^{-3} of total ionized impurity.
Excess carriers				
28	20-21 1(b)	5	Minority carriers	Minority carrier: the less numerous carrier ($p_{n0} = n_i^2/N_d$). It diffuses because random thermal motion in a concentration gradient gives a net flow from high to low; injected minority carriers recombine as they spread, so their current is diffusion current. Poisson: $d^2\phi/dx^2 = -\rho/\epsilon_s$ links the space charge to the field and potential, giving E_{max}, V_{bi} and W of a junction.
29	20-21 5(a)	5	Carrier pulse	For n-type at low injection the ambipolar equation uses D_p, μ_p, τ_{p0}: $D_p \, \partial^2 \delta p/\partial x^2 - \mu_p E_0 \, \partial \delta p/\partial x - \delta p/\tau_{p0} = \partial \delta p/\partial t$. A pulse at $x = 0$ becomes a Gaussian that spreads, shrinks by recombination and drifts in the direction of E_0 at $\mu_p E_0$.
30	20-21 TT1 7(c)	1	Excess carriers	n_0, p_0: thermal-equilibrium concentrations, $n_0 p_0 = n_i^2$; $n = n_0 + \delta n$, $p = p_0 + \delta p$: total concentrations including excess carriers from light or injection, may vary with time and position.
31	19-20 4(d)	6	Time and space dependence	Uniform: $\delta p = \delta p(0) e^{-t/\tau_{p0}}$ after the light stops, $g' \tau_{p0}(1 - e^{-t/\tau_{p0}})$ after it starts; steady injection: $\delta p = \delta p(0) e^{-x/L_p}$, $L_p = \sqrt{D_p \tau_{p0}}$; a pulse is a Gaussian that spreads ($4D_p t$), decays ($e^{-t/\tau_{p0}}$) and drifts at $\mu_p E_0$.
32	19-20 TT2 8(c)	7	Generation and recombination	Band to band: $h\nu \geq E_g$ or heat makes a pair, an electron falling into a hole emits a photon, rate $\alpha_r(np - n_i^2)$, GaAs. SRH: traps in the gap capture and emit electrons and holes (four processes), $R = C_n C_p N_t (np - n_i^2)/[C_n(n + n') + C_p(p + p')]$, non-radiative, dominant in Si. Auger: energy to a third carrier, heavy doping; inverse is impact ionization. Surface recombination through dangling-bond traps.
PN junction				
33	20-21 3(b)	5	Band diagram	E_F flat through the device; on the P side E_F is $e\phi_{Fp} = kT \ln(N_a/n_i)$ below E_{Fi}, on the N side $e\lvert\phi_{Fn}\rvert = kT \ln(N_d/n_i)$ above it; the bands are higher on the P side by eV_{bi} and bend only inside the space charge region; $V_{bi} = \lvert\phi_{Fp}\rvert + \lvert\phi_{Fn}\rvert$.
34	19-20 3(b)	4	Built-in potential	Diffusion uncovers ions, giving a field from n to p and a barrier $V_{bi} = (kT/e) \ln(N_a N_d / n_i^2)$; drift cancels diffusion for each carrier ($J_n = J_p = 0$); only $e^{-eV_{bi}/kT}$ of the majority carriers cross, $n_{p0} = n_{n0} e^{-eV_{bi}/kT}$; E_F flat.

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No.	Paper	Marks	Subtopic	Points to cover
35	18-19 2(a)	7	Switching transient	Forward bias stores excess minority charge $Q_p = I_F\tau_{p0}$ near the junction; on switching to reverse a constant I_R flows while the charge is removed (storage time t_s, $\text{erf}\sqrt{t_s/\tau_{p0}} = I_F/(I_F + I_R)$), then the current decays (fall time). Total reverse recovery time limits switching speed; cut it with short lifetime (gold doping), small I_F, large I_R, Schottky diodes.
36	18-19 2(d)	6	Junction capacitance	Reverse bias widens the space charge region; a change dV_R uncovers charge dQ' at its edges, so $C' = dQ'/dV_R = \epsilon_s/W = \{e\epsilon_s N_a N_d/[2(V_{bi} + V_R)(N_a + N_d)]\}^{1/2}$. It falls as $1/\sqrt{V_{bi} + V_R}$; $(1/C')^2$ is linear in V_R (intercept $-V_{bi}$, slope gives N_d). Varactor use.
37	18-19 4(a)	7	Ideal diode current	Minority diffusion currents $J_p(x_n)$ and $J_n(-x_p)$ at the space charge edges decay as $e^{-(x-x_n)/L_p}$ into the neutral regions; the majority currents rise to keep the total J constant everywhere; no recombination in the space charge region.
Tunnel diode				
38	20-21 5(c)	5	Operation	Both sides degenerately doped, so E_F lies inside the bands and the depletion layer is only 5 to 17 nm thick; electrons tunnel where filled states face empty ones; the current rises to a peak I_p, falls to a valley I_v (negative resistance), then rises as normal diode current; reverse bias tunnels freely.
39	18-19 4(e)	6	Operation	Both sides degenerately doped, very thin junction: electrons tunnel between filled n-side states and empty p-side states. Small forward bias: current rises to a peak; more bias: bands uncross, current falls (negative resistance) to a valley; then normal diffusion current. Reverse bias: large tunnelling current.
Metal-semiconductor contacts				
40	19-20 1(d)	6	Schottky and ohmic contacts	Metal on n-type with $\phi_m > \phi_s$: electrons leave the semiconductor, a depletion layer and upward band bending form a barrier $\phi_{B0} = \phi_m - \chi$, $V_{bi} = \phi_{B0} - \phi_n$; current by thermionic emission $J = J_{sT}(e^{eV_a/kT} - 1)$, rectifying, majority carriers, fast. Ohmic contact: $\phi_m < \phi_s$ (accumulation, no barrier) or a heavily doped surface so electrons tunnel; linear I-V, negligible resistance.
41	18-19 4(c)	4	Schottky and ohmic contacts	Rectifying (Schottky): $\phi_m > \phi_s$ on n-type, barrier $\phi_{B0} = \phi_m - \chi$, current by thermionic emission one way only, I-V like a diode. Ohmic: low or negative barrier ($\phi_m < \phi_s$ on n-type) or heavy doping so electrons tunnel through a thin barrier; linear I-V, small resistance either way.
42	18-19 6(d)	6	Schottky and ohmic contacts	Metal on n-type with $\phi_m > \phi_s$: electrons flow to the metal until E_F is flat, leaving a depletion region and barrier $\phi_{B0} = \phi_m - \chi$, $V_{bi} = \phi_{B0} - \phi_n$. Forward bias lowers V_{bi}: thermionic emission $J = A^*T^2 e^{-e\phi_{B0}/kT}(e^{eV_a/kT} - 1)$; reverse bias raises it. Majority carriers, no storage, fast switching; Ex 9.1: 0.54 V barrier.
MOS structure				
43	20-21 4(b)	5	Band diagrams	MOS on p-type Si. Accumulation ($V_G < 0$): bands bend up, holes gather at the surface. Depletion ($0 < V_G < V_T$): bands bend down, acceptor ions uncovered over x_d. Inversion ($V_G > V_T$): E_{Fi} at the surface falls below E_F, electrons form an n-type channel, $\phi_s = 2\phi_{fp}$ at threshold.
44	20-21 6(a)	5	Double gate	A thin (undoped) silicon body between two thin gate oxides with a gate on each side, n^+ source and drain (FinFET is its practical form). Benefits: better gate control of the channel, much smaller short-channel effects (DIBL, V_T roll-off), near-ideal subthreshold swing, low leakage, more drive current, no doping fluctuation.

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No.	Paper	Marks	Subtopic	Points to cover
45	20-21 6(c)	5	Inversion	Positive gate voltage on a p-type body repels holes and bends the bands down by $e\phi_s$: depletion for $\phi_s < \phi_{fp}$, intrinsic surface at $\phi_s = \phi_{fp}$, n-type inversion layer once E_{Fi} falls below E_F; threshold at $\phi_s = 2\phi_{fp}$ ($n_s = N_a$), depletion width at its maximum x_{dT}.
46	19-20 6(c)	3	MOSFET advantages	Insulated gate: very high input impedance, voltage control, almost no drive power; CMOS draws almost no static power; small, simple, dense; majority carrier, no storage delay; mobility falls with temperature, no thermal runaway, easy paralleling; good bidirectional switch.
47	18-19 3(a)	8	Band diagrams	Ideal MOS on p-type: $V_G = 0$ flat bands (flat band). $V_G < 0$: holes attracted, bands bend up, accumulation. Small $V_G > 0$: holes pushed away, bands bend down, depletion region. Large $V_G > 0$: E_{Fi} crosses E_F at the surface, electron inversion layer; threshold at $\phi_s = 2\phi_{fp}$.
48	18-19 3(b)	6	Threshold voltage	$V_{TN} = (\lvert Q'_{SD}(\max)\rvert - Q'_{ss})t_{ox}/\epsilon_{ox} + \phi_{ms} + 2\phi_{fp}$: substrate doping (through ϕ_{fp} and Q'_{SD}), oxide thickness and permittivity, fixed oxide charge Q'_{ss}, gate material (work function difference ϕ_{ms}), temperature, body bias (body effect), ion implant adjustment.
49	18-19 3(c)	5	Body effect	A reverse source-to-body bias V_{SB} widens the depletion region under the gate, adding bulk charge, so more gate voltage is needed: $\Delta V_T = \gamma(\sqrt{2\phi_{fp} + V_{SB}} - \sqrt{2\phi_{fp}})$, $\gamma = \sqrt{2e\epsilon_s N_a}/C_{ox}$ (body-effect coefficient). Matters in stacked transistors.
50	18-19 5(b)	5	C-V characteristics	p-type MOS: accumulation ($V_G < V_{FB}$): $C = C_{ox}$; depletion: C_{ox} in series with $C'_{SD} = \epsilon_s/x_d$, falls; minimum C'_{min} at threshold ($x_d = x_{dT}$); low-frequency inversion: inversion charge follows, C returns to C_{ox} (high frequency stays at C_{min}).
Bipolar transistor				
51	20-21 4(c)	5	Operating modes	Forward active ($V_{BE} > 0$, $V_{CB} > 0$), saturation (both junctions forward), cutoff (both reverse), inverse active; a forward junction raises the minority edge value by e^{V/V_t}, a reverse one takes it to 0; draw the base, emitter and collector minority profiles for each mode with the edge values.
52	20-21 5(b)	5	Minority distribution	Forward active: base electrons fall almost linearly from $n_{B0}e^{V_{BE}/V_t}$ at the emitter edge to about 0 at the collector edge; emitter holes fall from $p_{E0}e^{V_{BE}/V_t}$ to p_{E0}; collector holes rise from about 0 to p_{C0}.
53	19-20 6(d)	4	Emitter efficiency	Yes: $\gamma = J_{nE}/(J_{nE} + J_{pE}) \approx 1/[1 + (N_B/N_E)(D_E/D_B)(x_B/x_E)]$; heavier base doping lowers $n_{B0} = n_i^2/N_B$, so fewer electrons are injected into the base relative to holes injected back into the emitter, and γ falls; hence $N_E \gg N_B$.
54	18-19 3(d)	6	Amplification	Forward-biased B-E junction injects electrons that cross the thin base and are swept into the collector; $i_C = I_S e^{v_{BE}/V_t}$, so a small v_{be} makes a large change in i_C; through R_C this gives $v_{ce} = -g_m R_C v_{be}$, voltage gain $A_v = -I_C R_C/V_t$; current gain $\beta = i_C/i_B$.
55	18-19 4(b)	3	Base current	Holes injected from the base into the emitter (I_{pE}, emitter injection efficiency); recombination of electrons in the base (I_{RB}, base transport factor); recombination in the B-E space charge region (I_R); minus the reverse B-C leakage I_{CBO}.
56	18-19 4(d)	5	Base width modulation	Raising V_{CE} widens the B-C space charge region into the base, so the neutral base width x_B shrinks; the minority gradient and $I_C \propto 1/x_B$ rise; output curves slope and meet at $-V_A$ (Early voltage); also called the Early effect; finite output resistance.
57	18-19 6(c)	7	Current crowding	Base current flows sideways under the emitter through the base resistance; the $I_B r_b$ drop makes V_{BE} largest at the emitter edges, and injection $\propto e^{eV_{BE}/kT}$ crowds to the edges (18 mV doubles it). In power devices: local high current density, hot spots, second breakdown, lower gain; cured with narrow interdigitated emitters and ballast resistors.

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6. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
Semiconductor in equilibrium				
1	21-22 7(c)	5	Carrier concentrations	$n(E) = g_c f_F$; integrate from E_c; Boltzmann f_F; put in $g_c \propto \sqrt{E - E_c}$; $\eta = (E - E_c)/kT$; $\int \eta^{1/2} e^{-\eta} = \sqrt{\pi}/2$; $n_0 = N_c e^{-(E_c - E_F)/kT}$; same for holes; add and subtract E_{Fi}: $n_0 = n_i e^{(E_F - E_{Fi})/kT}$, $p_0 = n_i e^{-(E_F - E_{Fi})/kT}$; product $n_0 p_0 = n_i^2$.
2	20-21 3(a)	5	Carrier concentrations	Neutrality $n_0 + N_a = p_0 + N_d$ (complete ionization); put $p_0 = n_i^2/n_0$; multiply by n_0: quadratic; quadratic formula; keep the positive root; for p_0 put $n_0 = n_i^2/p_0$ instead.
3	20-21 6(b)	5	Ionization	Donor occupation with the factor $\frac{1}{2}$; non-degenerate: $n_d \approx 2N_d e^{-(E_d - E_F)/kT}$; $n_0 = N_c e^{-(E_c - E_F)/kT}$; the ratio cancels E_F; fraction $1/[1 + (N_c/2N_d)e^{-(E_c - E_d)/kT}]$; 0.4 % for P in Si at 10^{16} cm^{-3}, 300 K.
4	20-21 TT1 7(f)	5	Carrier concentrations	$n(E) = g_c f_F$; integrate over the band with the Boltzmann form; $\eta = (E - E_c)/kT$ and $\int_0^\infty \eta^{1/2} e^{-\eta} d\eta = \sqrt{\pi}/2$ give $n_0 = N_c e^{-(E_c - E_F)/kT}$; same with $1 - f_F$ for $p_0 = N_v e^{-(E_F - E_v)/kT}$; add and subtract E_{Fi}: $n_0 = n_i e^{(E_F - E_{Fi})/kT}$, $p_0 = n_i e^{-(E_F - E_{Fi})/kT}$; product $n_0 p_0 = n_i^2$.
5	20-21 TT2 8(f)	5	Carrier concentrations	Neutrality $n_0 + N_a = p_0 + N_d$ (complete ionization); put $p_0 = n_i^2/n_0$; multiply by n_0: quadratic; quadratic formula; keep the positive root; for p_0 put $n_0 = n_i^2/p_0$ instead.
6	19-20 2(a)	6	Carrier concentrations	Neutrality $n_0 + N_a = p_0 + N_d$ (complete ionization); put $p_0 = n_i^2/n_0$; multiply by n_0: quadratic; quadratic formula; keep the positive root; for p_0 put $n_0 = n_i^2/p_0$ instead.
7	19-20 TT1 7(a)	5	Carrier concentrations	Neutrality $n_0 + N_a = p_0 + N_d$ (complete ionization); put $p_0 = n_i^2/n_0$; multiply by n_0: quadratic; quadratic formula; keep the positive root; for p_0 put $n_0 = n_i^2/p_0$ instead.
Carrier transport				
8	20-21 2(b)	5	Einstein relation	Graded n-type sample in equilibrium: $J_n = 0$; $n = N_c e^{-(E_c - E_F)/kT}$ with E_F flat; chain rule gives $dn/dx = -(n/kT) dE_c/dx$; $dE_c/dx = eE_x$; substitute: $D_n/\mu_n = kT/e$; repeat for holes.
9	18-19 1(d)	7	Hall effect	Current I_x in field B_z: Lorentz force $q\mathbf{v} \times \mathbf{B}$ pushes carriers to $y = 0$; charge builds a Hall field until $eE_H = ev_x B_z$; $V_H = E_H W = v_x W B_z$; $v_x = I_x/(epWd)$; $V_H = I_x B_z/(epd)$, so $p = I_x B_z/(edV_H)$; electrons: $n = -I_x B_z/(edV_H)$; sign of V_H gives the type; mobility $\mu = I_x L/(epV_x Wd)$.
10	18-19 2(c)	6	Einstein relation	Non-uniformly doped n-type in equilibrium: $J_n = en\mu_n E_x + eD_n\, dn/dx = 0$; $E_x = (1/e) dE_c/dx$; $n = N_c e^{-(E_c - E_F)/kT}$ with flat E_F gives $dn/dx = -(en/kT)E_x$; substitute: $\mu_n = eD_n/kT$, so $D_n/\mu_n = kT/e$; same for holes.
11	18-19 5(a)	7	Diffusion	Carriers move randomly with v_{th} over mean free path l; half of those at $\pm l$ cross $x = 0$: $F = \frac{1}{2} v_{th}[n(-l) - n(+l)]$; Taylor $n(\pm l) = n(0) \pm l\, dn/dx$; $F = -v_{th} l\, dn/dx$; $J_n = -eF = eD_n\, dn/dx$ with $D_n = v_{th} l$; holes $J_p = -eD_p\, dp/dx$; total with drift.
Excess carriers				
12	19-20 3(c)	7	Dielectric relaxation	Continuity $\nabla \cdot J = -\partial\rho/\partial t$; Ohm $J = \sigma E$; Poisson $\nabla \cdot E = \rho/\epsilon$; $d\rho/dt + (\sigma/\epsilon)\rho = 0$; $\rho = \rho(0) e^{-t/\tau_d}$, $\tau_d = \epsilon/\sigma$ (0.539 ps for 10^{16} cm^{-3} Si).
13	19-20 4(b)	7	Ambipolar transport	Continuity for holes and electrons; drift-diffusion fluxes; time-dependent diffusion equations; excess form with $\delta n = \delta p$, $g_n = g_p$, $R_n = R_p$; multiply by $\mu_n n$ and $\mu_p p$ and add so $\partial E/\partial x$ cancels; divide: D', μ'; low injection: $D' = D_n$, $\mu' = \mu_n$ in p-type (D_p, $-\mu_p$ in n-type), so the majority carriers follow the minority carriers.

No.	Paper	Marks	Derivation	Outline
14	19-20 5(a)	7	Steady state	Ambipolar equation with minority electron parameters; $E = 0$, $g' = 0$ for $x \neq 0$, steady state; $\delta n'' - \delta n/L_n^2 = 0$; $Ae^{-x/L_n} + Be^{x/L_n}$; finite at $\pm\infty$; $\delta n = \delta n(0)e^{-\lvert x\rvert/L_n}$.
15	19-20 TT2 8(a)	6	Steady state	Ambipolar equation with minority electron parameters; $E = 0$, $g' = 0$ for $x \neq 0$, steady state; $\delta n'' - \delta n/L_n^2 = 0$; $Ae^{-x/L_n} + Be^{x/L_n}$; finite at $\pm\infty$; $\delta n = \delta n(0)e^{-\lvert x\rvert/L_n}$.
16	19-20 TT2 8(b)	7	Ambipolar transport	Continuity for holes and electrons; drift-diffusion fluxes; time-dependent diffusion equations; excess form with $\delta n = \delta p$, $g_n = g_p$, $R_n = R_p$; multiply by $\mu_n n$ and $\mu_p p$ and add so $\partial E/\partial x$ cancels; divide: D', μ'; low injection: $D' = D_n$, $\mu' = \mu_n$ in p-type (D_p, $-\mu_p$ in n-type), so the majority carriers follow the minority carriers.
17	18-19 5(c)	6	Steady state	Steady state, no field: $D_p\, d^2\delta p/dx^2 - \delta p/\tau_{p0} = 0$; $L_p = \sqrt{D_p\tau_{p0}}$; $\delta p = \delta p_0 e^{-x/L_p}$; probability of recombining in dx: $P(x)dx = (1/L_p)e^{-x/L_p}dx$; mean $\langle x\rangle = (1/L_p)\int_0^\infty xe^{-x/L_p}dx = L_p$ (integration by parts).
PN junction				
18	20-21 2(c)	0	Depletion width	Charge density $-eN_a$, $+eN_d$; integrate Poisson once with $E = 0$ at both edges; continuity gives $N_a x_p = N_d x_n$; integrate again for the potential; $V_{bi} = \phi(x_n)$; substitute $x_p = N_d x_n/N_a$; solve for x_n, x_p; add for W.
19	19-20 5(b)	8	Ideal diode current	$n_{p0} = n_{n0}e^{-eV_{bi}/kT}$; barrier $V_{bi} - V_a$ gives the law of the junction $p_n(x_n) = p_{n0}e^{eV_a/kT}$, $n_p(-x_p) = n_{p0}e^{eV_a/kT}$; minority diffusion equations; decaying exponentials; diffusion currents at the edges; add: $J = [eD_p p_{n0}/L_p + eD_n n_{p0}/L_n](e^{eV_a/kT} - 1)$; majority currents make up the rest.

7. Numericals:

Every numerical asked, in book order: the method in one line and the final answer, so a worked attempt can be checked. The link opens the full working.

No.	Paper	Marks	Subtopic	Method	Answer
Crystal structure					
1	19-20 1(b)	6	Unit cell	fcc: 8 corner atoms shared by 8 cells, 6 face atoms shared by 2: $8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$; volume density $= 4/a^3$ with a in cm.	(i) 4 atoms per unit cell; (ii) 5.21×10^{22} cm^{-3} ($a = 0.425$ nm)
Quantum basics					
2	21-22 7(d)	5	Fermi probability	Set $1/[1 + e^{(E-E_F)/kT}] = 0.1$; $e^{(E-E_F)/kT} = 9$; $E - E_F = kT\ln 9$ (do not use the Boltzmann form).	$E - E_F = 2.197\,kT \approx 2.2\,kT$ (0.0569 eV at 300 K)
3	20-21 1(c)	5	Density of states	Integrate the density of states $g_c(E) = 4\pi(2m_n^*)^{3/2}\sqrt{E - E_c}/h^3$ from E_c to $E_c + kT$: $N = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \cdot \frac{2}{3}(kT)^{3/2}$.	$N = 2.13 \times 10^{19}$ cm^{-3}
4	20-21 2(d)	10	Fermi probability	(i) Set $f_F = 1/[1 + e^{(E-E_F)/kT}] = 0.1$ and solve: $E - E_F = kT\ln 9$. (ii) $n_0 = N_c \exp[-(E_c - E_F)/kT]$.	(i) $2.2\,kT$ (0.0569 eV); (ii) 1.16×10^{11} cm^{-3}
5	20-21 TT2 8(g)	5	Fermi probability	Set $f_F = 1/[1 + e^{(E-E_F)/kT}] = 0.1$ and solve: $E - E_F = kT\ln 9$.	$2.197\,kT \approx 2.2\,kT$ (0.0569 eV)
6	20-21 TT2 8(h)	5	Fermi probability	$n_0 = N_c \exp[-(E_c - E_F)/kT]$ with $E_c - E_F = 0.5$ eV.	$n_0 = 1.16 \times 10^{11}$ cm^{-3}

No.	Paper	Marks	Subtopic	Method	Answer
7	19-20 TT1 7(d)	5	Fermi probability	$f_F(E_c) = 1/[1 + e^{(E_c - E_F)/kT}]$; $n_0 = N_c e^{-(E_c - E_F)/kT}$; $p_0 = n_i^2/n_0$.	(i) 9.32×10^{-6}; (ii) $n_0 = 2.61 \times 10^{14}$ cm^{-3}; (iii) $p_0 = 8.62 \times 10^5$ cm^{-3}
Semiconductor in equilibrium					
8	21-22 7(e)	0	Carrier concentrations	$n_0 = N_c \exp[-(E_c - E_F)/kT]$ with $E_c - E_F = 0.5$ eV $= 19.3\,kT$.	$n_0 = 1.16 \times 10^{11}$ cm^{-3}
9	21-22 7(f)	7	Compensation	Si on Ga sites are donors, on As sites acceptors: $N_d = 0.05N$, $N_a = 0.95N$; $N_a > N_d$ so p-type; $p_0 = (N_a - N_d)/2 + \sqrt{((N_a - N_d)/2)^2 + n_i^2}$, $n_0 = n_i^2/p_0$; $E_{Fi} - E_F = kT \ln(p_0/n_i)$.	$N_d = 3.5 \times 10^{14}$, $N_a = 6.65 \times 10^{15}$ cm^{-3}; p-type; $p_0 = 6.3 \times 10^{15}$, $n_0 = 5.14 \times 10^{-4}$ cm^{-3}; E_F 0.569 eV below E_{Fi}
10	20-21 1(d)	10	Ionization	$n_0 = N_d$, $p_0 = n_i^2/n_0$; $E_c - E_F = kT \ln(N_c/n_0)$, $E_F - E_v = E_g - (E_c - E_F)$; fraction on donors $1/[1 + (N_c/2N_d)e^{-(E_c - E_d)/kT}]$.	$n_0 = 10^{15}$, $p_0 = 2.25 \times 10^5$ cm^{-3}; $E_c - E_F = 0.2652$ eV, $E_F - E_v = 0.8548$ eV; 99.95 % ionized
11	20-21 TT1 7(g)	5	Carrier concentrations	Neutrality with $p_0 = n_i^2/n_0$: $n_0 = N_d/2 + \sqrt{(N_d/2)^2 + n_i^2}$; with $N_d \gg n_i$, $n_0 = N_d$ (printed 10^6 read as 10^{16}).	$n_0 = 1 \times 10^{16}$ cm^{-3}
12	20-21 TT1 7(h)	5	Carrier concentrations	Mass-action law $p_0 = n_i^2/n_0$.	$p_0 = 2.25 \times 10^4$ cm^{-3}
13	19-20 TT1 7(c)	5	Carrier concentrations	Neutrality and mass action: $n_i^2 = n_0(n_0 - N_d)$ at $n_0 = 1.01 \times 10^{15}$; $E_g = kT \ln(N_c N_v/n_i^2)$ with $kT = 0.0259 \times 400/300$.	$n_i = 1.005 \times 10^{14}$ cm^{-3}; $E_{g,min} = 0.823$ eV
14	18-19 1(b)	6	Carrier concentrations	Neutrality $p_0 = n_0 + N_a$ with $n_0 = n_i^2/p_0$: $p_0 = N_a/2 + \sqrt{(N_a/2)^2 + n_i^2}$; then $n_0 = n_i^2/p_0$.	$p_0 = 1.0006 \times 10^{15}$ cm^{-3}, $n_0 = 6.25 \times 10^{11}$ cm^{-3}
15	18-19 TT2 8(b)	4	Fermi level	Compensated p-type: $p_0 = N_a - N_d$; $E_F - E_v = kT \ln(N_v/p_0)$ with $N_v = 7.0 \times 10^{18}$ cm^{-3} for GaAs.	$p_0 = 4.6 \times 10^{16}$ cm^{-3}; $E_F - E_v = 0.130$ eV
Carrier transport					
16	20-21 5(d)	10	Hall effect	$V_H = V_y$ from the figure; its sign gives the carrier type; $n = -I_x B_z/(edV_H)$ with d the thickness along z; $\mu_n = I_x L/(enV_x W d)$.	$V_H = -6.25$ mV; n-type (electrons); $n = 1.0 \times 10^{16}$ cm^{-3}; $\mu_n = 833$ cm^2/V s
17	19-20 1(c)	7	Thermal velocity	Maxwell-Boltzmann speed distribution; mean speed $\bar{v} = \sqrt{8kT/\pi m^*}$ (rms $\sqrt{3kT/m^*}$ for comparison) with $m^* = 0.1m_0$ and $1.5m_0$.	Electrons 3.40×10^5 m/s, holes 8.78×10^4 m/s (rms 3.69×10^5, 9.53×10^4 m/s)
18	19-20 2(b)	8	Hall effect	$V_H = E_H W$; its sign gives the carrier type; $n = -I_x B_z/(edV_H)$ with d in metres; $\mu_n = I_x L/(enV_x W d)$.	$V_H = -0.825$ mV; n-type; $n = 4.92 \times 10^{15}$ cm^{-3}; $\mu_n = 1015$ cm^2/V s
19	19-20 3(d)	6	Mobility	$\sigma = e\mu_n(N_d - N_a)$ with μ_n read at $N_I = N_d + N_a$ from the mobility curve; trial and error (ρ printed for σ).	$N_d \approx 3.5 \times 10^{17}$ cm^{-3}, $\mu_n \approx 400$ cm^2/V s

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No.	Paper	Marks	Subtopic	Method	Answer
20	18-19 TT2 8(c)	3	Mobility	Mobility from the total $N_I = N_d + N_a$ (Neamen Fig. 5.3); $n_0 = N_d - N_a$; $\sigma = e(\mu_n n_0 + \mu_p p_0) \approx e\mu_n n_0$; $\rho = 1/\sigma$.	$\mu_n \approx 1000$, $\mu_p \approx 350$ cm^2/V s; $\sigma = 4.8$ $(\Omega\text{-cm})^{-1}$, $\rho = 0.208$ Ω-cm
Excess carriers					
21	20-21 3(d)	10	Steady state	Low injection in p-type: ambipolar equation becomes $D_n\,\delta n'' - \delta n/\tau_{n0} = 0$; solution decaying for $x > 0$: $\delta n = \delta n_0 e^{-x/L_n}$, $L_n = \sqrt{D_n \tau_{n0}}$; solve $\delta n(x) = 10^{14}$ for x.	$\delta n = 5 \times 10^{15} e^{-x/L_n}$, $L_n = 45.8$ μm; 10^{14} cm^{-3} at 179.3 μm
22	19-20 5(c)	4	Quasi-Fermi levels	$E_{Fn} - E_{Fi} = kT \ln[(n_0 + \delta n)/n_i]$, $E_{Fi} - E_{Fp} = kT \ln[(p_0 + \delta p)/n_i]$ with $\delta n = \delta p = 10^{13}$ cm^{-3} (printed 10^{-13}).	$E_{Fn} - E_{Fi} = 0.2984$ eV, $E_{Fi} - E_{Fp} = 0.179$ eV ($E_F - E_{Fi} = 0.2982$ eV)
PN junction					
23	20-21 4(d)	10	Biased junction	$p_{n0} = n_i^2/N_d$; law of the junction $p_n(x_n) = p_{n0} e^{eV_a/kT}$ with $V_a = +0.7$ and -0.7 V; $n_n = N_d + \delta p$ (neutrality); check low injection.	$p_{n0} = 2.25 \times 10^4$; forward: $p_n = 1.23 \times 10^{16}$, $n_n = 2.23 \times 10^{16}$ cm^{-3} (low injection violated); reverse: $p_n \approx 0$, $n_n \approx 10^{16}$ cm^{-3}
24	20-21 6(d)	10	Depletion width	$V_{bi} = V_t \ln\!\left(N_a N_d / n_i^2\right)$; $W = \{(2\epsilon_s V_{bi}/e)[(N_a + N_d)/(N_a N_d)]\}^{1/2}$; $x_n = W N_a/(N_a + N_d)$; $E_{max} = -eN_d x_n/\epsilon_s$.	$V_{bi} = 0.718$ V; $W = 0.452$ μm; $E_{max} = -3.18 \times 10^4$ V/cm
25	19-20 4(c)	6	Depletion width	$V_{bi} = V_t \ln\!\left(N_a N_d / n_i^2\right)$; $W = \{(2\epsilon_s V_{bi}/e)[(N_a + N_d)/(N_a N_d)]\}^{1/2}$; $E_{max} = -eN_d x_n/\epsilon_s$; at 80 K scale n_i by $(T/300)^{3/2} e^{-E_g/2k(1/T - 1/300)}$ and V_t by $T/300$.	$V_{bi} = 0.820$ V, $W = 0.267$ μm, $E_{max} = -6.13 \times 10^4$ V/cm; 80 K: $V_{bi} = 1.069$ V, $W = 0.305$ μm
26	19-20 5(d)	6	Depletion width	$V_{bi} = V_t \ln\!\left(N_a N_d / n_i^2\right)$; $W = \{(2\epsilon_s (V_{bi} + V_R)/e)[(N_a + N_d)/(N_a N_d)]\}^{1/2}$; $\lvert E_{max}\rvert = 2(V_{bi} + V_R)/W$ at $V_R = 0$ and 8 V.	$V_{bi} = 0.671$ V; $W = 0.691$ and 2.48 μm; $E_{max} = -1.94 \times 10^4$ and -6.98×10^4 V/cm
27	18-19 TT2 8(a)	3	Depletion width	$V_{bi} = (kT/e) \ln\!\left(N_a N_d / n_i^2\right)$; $x_n = [(2\epsilon_s V_{bi}/e)(N_a/N_d)/(N_a + N_d)]^{1/2}$, x_p with N_d/N_a; $W = x_n + x_p$; check $N_a x_p = N_d x_n$.	$V_{bi} = 0.718$ V; $x_n = 0.0411$ μm, $x_p = 0.411$ μm, $W = 0.452$ μm
MOS structure					
28	19-20 6(a)	12	Threshold voltage	$\phi_{fp} = V_t \ln(N_A/n_i)$; $x_{dT} = (4\epsilon_s \phi_{fp}/eN_A)^{1/2}$; $\lvert Q'_{SD}\rvert = eN_A x_{dT}$; $\lvert Q'_I\rvert = \epsilon_{ox} E_{ox} - \lvert Q'_{SD}\rvert$; electrons $= \lvert Q'_I\rvert LW/e$; $V_T = V_{FB} + 2\phi_{fp} + \lvert Q'_{SD}\rvert/C_{ox}$; $V_G = V_T + \lvert Q'_I\rvert/C_{ox}$ (N_A printed 5^{18}, read 5×10^{18}).	$\lvert Q_I\rvert = 7.73 \times 10^{-7}$ C/cm^2; 1208 electrons; $V_T = 2.39$ V; $V_G = 2.62$ V

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No.	Paper	Marks	Subtopic	Method	Answer
Bipolar transistor					
29	19-20 6(b)	6	Operating modes	$V_{BC} = V_{BE} - V_{CE}$; B-E and B-C forward or reverse decide the mode; from currents compare I_C/I_B with β ($I_B = I_E - I_C$).	i forward active; ii saturation; iii saturation (soft); iv inverse active; v forward active; vi saturation ($I_C/I_B = 9.7$)
30	18-19 6(b)	6	Emitter efficiency	$\gamma = 1/[1 + (N_B/N_E)(D_E/D_B)(L_B/L_E)\tanh(x_B/L_B)/\tanh(x_E/L_E)]$; with equal D, L and x: $\gamma = 1/(1 + N_B/N_E)$, so $N_E/N_B = 1/(1/\gamma - 1)$.	$N_E/N_B = 302$
JFET					
31	18-19 5(d)	7	Pinch-off	$V_{bi} = V_t \ln(N_a N_d/n_i^2)$; $V_{p0} = V_{bi} - V_p$; $a = [2\epsilon_s V_{p0}/(eN_d)]^{1/2}$.	$V_{bi} = 0.832$ V, $V_{p0} = 3.33$ V, $a = 0.464$ μm
32	18-19 6(a)	6	Frequency response	$f_T = g_m/(2\pi C_G)$ with $g_m(\max) = e\mu_n N_d W a/L$ and $C_G(\min) = \epsilon_s WL/a$: $f_T = e\mu_n N_d a^2/(2\pi\epsilon_s L^2)$.	$f_T = 7.69$ GHz

8. Diagrams to practise:

Every figure an answer needs, whatever the question's type, in book order. Draw each one from memory, then check it against the solution.

No.	Paper	Kind	Diagram
Crystal structure			
1	19-20 1(a)	Diagram	atoms in amorphous, polycrystalline and single-crystal solids
2	19-20 1(b)	Diagram	fcc unit cell with corner and face atoms
Quantum basics			
3	21-22 7(a)	Graph	Fermi-Dirac function at 0 K, 300 K and 600 K
4	21-22 7(a)	Graph	f_F and $1 - f_F$ symmetric about E_F
5	21-22 7(d)	Graph	Fermi-Dirac function with the 10 % point
6	20-21 1(c)	Graph	density of states with the shaded strip
7	20-21 2(d)	Graph	Fermi-Dirac function with the 10 % point
8	20-21 TT2 8(a)	Graph	Fermi-Dirac function at three temperatures
9	20-21 TT2 8(g)	Graph	Fermi-Dirac function with the 10 % point
10	19-20 TT1 7(d)	Diagram	Si band diagram with E_F 0.3 eV below E_c
11	18-19 1(c)	Graph	Fermi-Dirac function at 0 K
12	18-19 1(c)	Diagram	full valence band and empty conduction band at 0 K
13	18-19 2(b)	Graph	Fermi-Dirac function with $f_F = 1/2$ at E_F
14	18-19 7(a)	Graph	density of states in the conduction band of Si
15	18-19 7(c)	Graph	E-k diagram of bands A and B (Figure 1)
16	18-19 7(c)	Graph	parabolic fits of bands A and B with their curvatures
17	18-19 7(d)	Diagram	splitting of atomic levels into bands as atoms come together
18	18-19 7(d)	Graph	Kronig-Penney function with allowed and forbidden bands
19	18-19 7(d)	Graph	E-k diagram from the Kronig-Penney model
Semiconductor in equilibrium			
20	21-22 7(b)	Diagram	donor electron orbiting its ion in the Si lattice
21	21-22 7(b)	Diagram	donor and acceptor ionization on the band diagram

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No.	Paper	Kind	Diagram
22	21-22 7(c)	Graph	density of states, Fermi function and electron and hole distributions for n-type and p-type
23	21-22 7(e)	Diagram	Si band diagram with E_F 0.5 eV below E_c
24	21-22 7(f)	Diagram	Si on Ga and As sites in GaAs
25	21-22 7(f)	Diagram	GaAs band diagram with E_F below E_{Fi}
26	20-21 1(a)	Graph	Fermi-Dirac function at three temperatures
27	20-21 1(a)	Diagram	band diagrams of intrinsic, n-type and p-type Si
28	20-21 1(a)	Graph	Fermi level against doping at 300 K
29	20-21 1(d)	Diagram	band diagram of the As-doped sample
30	20-21 1(d)	Graph	ionized fraction and Fermi level against temperature
31	20-21 2(a)	Diagram	band diagrams of a compensated semiconductor for three cases
32	20-21 2(a)	Graph	Fermi level against acceptor concentration
33	20-21 3(a)	Diagram	charge balance in a compensated semiconductor
34	20-21 6(b)	Diagram	donor states at 0 K, low temperature and 300 K
35	20-21 6(b)	Graph	n_0/N_d against temperature
36	20-21 TT1 7(a)	Diagram	donor states at 0 K, low temperature and 300 K
37	20-21 TT1 7(b)	Graph	Fermi-Dirac function and donor occupation function
38	20-21 TT1 7(d)	Diagram	Si on a Ga site, on an As site and in silicon
39	20-21 TT1 7(e)	Graph	n_0/N_d against temperature
40	20-21 TT1 7(f)	Graph	density of states, occupation and carrier distributions
41	20-21 TT2 8(b)	Diagram	band diagrams of a compensated semiconductor for three cases
42	20-21 TT2 8(c)	Diagram	E_c, E_v, E_{Fi} and E_F of p-type Si, E_F below E_{Fi} near E_v
43	20-21 TT2 8(d)	Diagram	Si on a Ga site, on an As site and in silicon
44	20-21 TT2 8(e)	Diagram	non-degenerate and degenerate n- and p-type band diagrams
45	20-21 TT2 8(f)	Diagram	charge balance in a compensated semiconductor
46	19-20 2(a)	Diagram	charge balance in a compensated semiconductor
47	19-20 2(c)	Graph	electron concentration against temperature with the three regions
48	19-20 2(d)	Diagram	band diagrams of a compensated semiconductor for three cases
49	19-20 3(a)	Graph	density of states, Fermi function and carrier distributions of n-type on one energy axis
50	19-20 TT1 7(a)	Diagram	charge balance in a compensated semiconductor
51	19-20 TT1 7(b)	Graph	Fermi level against doping at 300 K
52	19-20 TT1 7(b)	Graph	Fermi level against temperature for several dopings
53	19-20 TT1 7(c)	Graph	n_0 and n_i at 400 K against E_g
54	18-19 1(a)	Graph	electron concentration against temperature with freeze-out, extrinsic and intrinsic regions
55	18-19 1(a)	Graph	Fermi level against temperature
56	18-19 1(a)	Diagram	donor states at 0 K, low T and 300 K

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No.	Paper	Kind	Diagram
57	18-19 7(b)	Graph	density of states, Fermi function and their product near the conduction band
58	18-19 TT2 8(b)	Diagram	GaAs band diagram with the Fermi level near the valence band
Carrier transport			
59	20-21 2(b)	Diagram	graded doping profile and its equilibrium band diagram with the induced field
60	20-21 3(c)	Diagram	electron path with and without a field
61	20-21 3(c)	Diagram	lattice and ionized impurity scattering
62	20-21 4(a)	Graph	mobility against temperature, log-log
63	20-21 5(d)	Diagram	Hall bar with the force balance
64	20-21 5(d)	Diagram	carriers piling up in p-type and n-type bars
65	19-20 1(c)	Graph	Maxwell-Boltzmann speed distributions for the two masses
66	19-20 2(b)	Diagram	Hall bar with dimensions, current, field and Hall field
67	19-20 3(d)	Graph	electron mobility against total impurity with the trial points
68	19-20 4(a)	Graph	mobility against temperature, log-log
69	19-20 4(a)	Graph	electron and hole mobility against doping
70	18-19 1(d)	Diagram	Hall bar with current, magnetic field, Hall field and Hall voltage
71	18-19 2(c)	Diagram	band diagram of a graded n-type semiconductor with drift and diffusion
72	18-19 5(a)	Graph	electron concentration against distance with the flux across $x = 0$
73	18-19 TT2 8(c)	Graph	electron and hole mobility against total impurity concentration
Excess carriers			
74	20-21 1(b)	Diagram	diffusion of carriers down a concentration gradient
75	20-21 1(b)	Graph	charge density, field and potential of a PN junction
76	20-21 3(d)	Graph	excess electrons decaying into the bar
77	20-21 3(d)	Graph	total electron and hole concentrations on a log scale
78	20-21 5(a)	Diagram	excess electrons and holes held together by their field
79	20-21 5(a)	Graph	excess carrier pulse at three times, with and without a field
80	20-21 TT1 7(c)	Diagram	n-type sample in equilibrium and under light
81	19-20 3(c)	Diagram	net charge drawing in majority carriers
82	19-20 3(c)	Graph	exponential decay of the net charge
83	19-20 4(b)	Diagram	excess electrons and holes held together by their internal field
84	19-20 4(d)	Graph	time decay and build-up of a uniform excess
85	19-20 4(d)	Graph	steady-state decay from a surface
86	19-20 4(d)	Graph	excess carrier pulse at three times with and without a field
87	19-20 5(a)	Graph	excess carriers decaying on both sides of the source
88	19-20 5(c)	Diagram	band diagram in equilibrium and with quasi-Fermi levels
89	19-20 TT2 8(a)	Graph	excess carriers decaying on both sides of the source
90	19-20 TT2 8(b)	Diagram	excess electrons and holes held together by their internal field
91	19-20 TT2 8(c)	Diagram	band-to-band, trap (SRH) and Auger processes
92	18-19 5(c)	Graph	excess carrier concentration decaying with distance, with L_p marked
PN junction			
93	20-21 2(c)	Diagram	PN junction with its space charge region
94	20-21 2(c)	Graph	charge density, field and potential across an abrupt junction
95	20-21 3(b)	Diagram	p and n band diagrams before contact
96	20-21 3(b)	Diagram	equilibrium band diagram of a PN junction with all labels
97	20-21 4(d)	Diagram	band diagrams under forward and reverse bias

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No.	Paper	Kind	Diagram
98	20-21 4(d)	Graph	hole and electron concentrations on the n side
99	20-21 6(d)	Graph	charge density, field and potential across the junction
100	19-20 3(b)	Diagram	equilibrium band diagram of a pn junction
101	19-20 3(b)	Diagram	drift and diffusion fluxes across the space charge region
102	19-20 4(c)	Graph	space charge and field triangles at 300 K and 80 K
103	19-20 5(b)	Graph	minority carrier distributions on both sides of a forward-biased junction
104	19-20 5(b)	Graph	electron, hole and total current against position
105	19-20 5(d)	Graph	field triangles at 0 and 8 V reverse bias
106	18-19 2(a)	Graph	diode current and voltage against time when switched from forward to reverse
107	18-19 2(a)	Graph	stored minority charge removed during storage time
108	18-19 2(d)	Diagram	charge added at the edges of the space charge region for a change in reverse bias
109	18-19 2(d)	Graph	junction capacitance against reverse voltage
110	18-19 2(d)	Graph	$(1/C)^2$ against reverse voltage
111	18-19 4(a)	Graph	electron and hole current densities through a forward-biased pn junction
112	18-19 TT2 8(a)	Graph	charge density, electric field and potential across the junction
Tunnel diode			
113	20-21 5(c)	Diagram	electron wave at a thick and a thin barrier
114	20-21 5(c)	Diagram	tunnel diode band diagrams at several biases
115	20-21 5(c)	Graph	tunnel diode I-V curve
116	18-19 4(e)	Diagram	tunnel diode band diagrams at zero bias, peak, valley and reverse bias
117	18-19 4(e)	Graph	tunnel diode I-V characteristic with peak, valley and negative resistance
Metal-semiconductor contacts			
118	19-20 1(d)	Diagram	metal and n-type band diagrams before and after contact
119	19-20 1(d)	Diagram	ohmic contact band diagram
120	19-20 1(d)	Graph	I-V of rectifying and ohmic contacts
121	18-19 4(c)	Diagram	band diagrams of rectifying and ohmic contacts
122	18-19 4(c)	Graph	I-V curves of rectifying and ohmic contacts
123	18-19 6(d)	Diagram	metal and n-type band diagrams before and after contact
124	18-19 6(d)	Diagram	Schottky contact under forward and reverse bias
MOS structure			
125	20-21 4(b)	Diagram	MOS capacitor and n-channel MOSFET cross sections
126	20-21 4(b)	Diagram	charges in accumulation, depletion and inversion
127	20-21 4(b)	Diagram	band diagrams in the three modes
128	20-21 6(a)	Diagram	single-gate, planar double-gate and FinFET schematics
129	20-21 6(a)	Graph	subthreshold current of single and double gate
130	20-21 6(c)	Diagram	band diagrams of a p-type MOS structure from depletion to inversion
131	20-21 6(c)	Diagram	charge distribution at threshold inversion
132	19-20 6(a)	Diagram	MOS charges at the maximum inversion charge with the Gauss balance
133	18-19 3(a)	Diagram	MOS band diagrams at zero, negative, small positive and large positive gate voltage
134	18-19 3(b)	Graph	threshold voltage against substrate doping for several oxide thicknesses
135	18-19 3(c)	Graph	threshold voltage against source-to-body voltage
136	18-19 3(c)	Diagram	depletion region with and without body bias
137	18-19 5(b)	Graph	ideal low-frequency C-V curve of a p-type MOS capacitor with the three regions
138	18-19 5(b)	Diagram	charge in accumulation, depletion and inversion
Bipolar transistor			
139	20-21 4(c)	Diagram	NPN structure and the four modes on the V_{BE}–V_{CB} plane
140	20-21 4(c)	Graph	minority carrier distributions in the four modes
141	20-21 5(b)	Diagram	NPN in forward-active mode with depletion layers

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No.	Paper	Kind	Diagram
142	20-21 5(b)	Graph	minority carrier concentrations in emitter, base and collector
143	19-20 6(b)	Diagram	four modes on the V_{BE}–V_{BC} plane with the cases
144	19-20 6(d)	Diagram	minority carriers in emitter, base and collector
145	19-20 6(d)	Graph	γ against N_B/N_E
146	18-19 3(d)	Circuit	common-emitter amplifier
147	18-19 3(d)	Graph	input and output waveforms of the amplifier
148	18-19 4(b)	Diagram	current components in an npn transistor in active mode
149	18-19 4(d)	Graph	minority carrier distribution in the base for two collector voltages
150	18-19 4(d)	Graph	collector current against collector-emitter voltage with the Early voltage
151	18-19 6(b)	Graph	emitter injection efficiency against the doping ratio
152	18-19 6(c)	Diagram	npn cross section with lateral base current and injection crowded at the emitter edges
153	18-19 6(c)	Diagram	interdigitated emitter and base fingers
JFET			
154	18-19 5(d)	Diagram	n-channel JFET cross section with channel thickness and depletion regions

9. Comparisons:

Every comparison asked, in book order, with the key difference in one line. A pair asked in several papers is one row.

Pair	Key difference	Asked in
Crystal structure		
Amorphous vs polycrystalline vs single crystal	short-range order only vs ordered grains with boundaries vs one lattice throughout	19-20 1(a)
Quantum basics		
Density of states vs carrier concentration	states available vs states filled	18-19 7(a)
Band A vs band B	larger curvature, lighter mass vs flatter, heavier mass	18-19 7(c)
Fermi-Dirac vs Boltzmann	exact, at most one electron per state vs approximation valid for $E - E_F > 3kT$	21-22 7(a)
Semiconductor in equilibrium		
Freeze-out vs intrinsic region	dopants not ionized, $n_0 < N_d$ vs thermal generation, $n_0 \to n_i$	18-19 1(a)
Complete ionization vs freeze-out	every donor gives its electron, $n_0 = N_d$ vs electrons fall back to donors at low T, $n_0 < N_d$	20-21 6(b)
Degenerate vs non-degenerate	E_F inside a band, metal-like vs E_F in the gap, $n_0 p_0 = n_i^2$ holds	20-21 TT2 8(e)
Donor vs acceptor	gives an electron, level near E_c, becomes + vs takes an electron, level near E_v, becomes −	21-22 7(b)
Si on a Ga site vs Si on an As site	donor (+) vs acceptor (−)	21-22 7(f)
Carrier transport		
Net vs total doping	$N_d - N_a$ sets n_0 vs $N_d + N_a$ sets the mobility	18-19 TT2 8(c)
Lattice vs ionized impurity scattering	grows with temperature, $\mu_L \propto T^{-3/2}$ vs weakens with temperature, $\mu_I \propto T^{3/2}/N_I$	20-21 3(c), 19-20 4(a)
Drift vs diffusion current	driven by a field, $en\mu E$ vs driven by a gradient, $eD\,dn/dx$; linked by $D/\mu = kT/e$	20-21 2(b)
Excess carriers		
Band-to-band vs SRH recombination	direct, radiative, GaAs vs through traps, heat, Si	19-20 TT2 8(c)

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Pair	Key difference	Asked in
Minority vs majority carriers	fewer, set by n_i^2/N, move by diffusion vs more, set by the doping, carry drift current	20-21 1(b)
Equilibrium vs total concentration	n_0 fixed by doping and T vs $n = n_0 + \delta n$ out of equilibrium	20-21 TT1 7(c)
PN junction		
Storage time vs fall time	stored charge removed at constant I_R vs current decays to 10 %	18-19 2(a)
Minority vs majority current	diffusion, decays over L vs makes up the constant total	18-19 4(a)
Metal-semiconductor contacts		
Ohmic vs rectifying contact	linear, either direction vs barrier, one direction	18-19 4(c)
Schottky vs pn diode	majority carriers, no stored charge, lower turn-on vs minority injection, storage time	18-19 6(d)
Schottky barrier vs ohmic contact	depletion, barrier, thermionic emission, rectifying vs accumulation or tunneling, no barrier, linear	19-20 1(d)
MOS structure		
Accumulation vs depletion vs inversion	majority carriers gather vs carriers pushed away, ions uncovered vs minority carriers form a channel	20-21 4(b)
Single-gate vs double-gate MOSFET	gate controls one surface, short-channel effects vs gate on both sides, better control, near-ideal swing	20-21 6(a)

10. Drills:

Every drawing drill asked, in book order: what to draw and the data given. The data changes from paper to paper; the method does not.

Draw	Data given	Asked in
Semiconductor in equilibrium		
p-type band diagram to scale	$N_a = 10^{16}$ cm^{-3}, $E_{Fi} - E_F = 0.35$ eV, $E_F - E_v = 0.18$ eV	20-21 TT2 8(c)

11. Repeated questions:

14 questions came back, exactly or with new numbers; 2 of them in three or more papers. Most repeated first; the latest paper first in each row.

No.	Times	Topic	Subtopic	Asked in (marks)
1	6	Semiconductor in equilibrium	Carrier concentrations	21-22 7(c) (5), 20-21 3(a) (5), 20-21 TT1 7(f) (5), 20-21 TT2 8(f) (5), 19-20 2(a) (6), 19-20 TT1 7(a) (5)
2	6	Quantum basics	Fermi probability	21-22 7(d) (5), 21-22 7(e), 20-21 2(d) (10), 20-21 TT1 7(h) (5), 20-21 TT2 8(g) (5), 20-21 TT2 8(h) (5)
3	2	Carrier transport	Einstein relation	20-21 2(b) (5), 18-19 2(c) (6)
4	2	MOS structure	Band diagrams	20-21 4(b) (5), 18-19 3(a) (8)
5	2	Tunnel diode	Operation	20-21 5(c) (5), 18-19 4(e) (6)
6	2	PN junction	Depletion width	20-21 6(d) (10), 18-19 TT2 8(a) (3)
7	2	Semiconductor in equilibrium	Compensation	20-21 2(a) (5), 19-20 2(d) (6)
8	2	Carrier transport	Mobility	20-21 4(a) (5), 19-20 4(a) (6)
9	2	Semiconductor in equilibrium	Dopants	20-21 TT1 7(d) (1), 20-21 TT2 8(d) (1)
10	2	Semiconductor in equilibrium	Temperature dependence	19-20 2(c) (5), 18-19 1(a) (7)

No.	Times	Topic	Subtopic	Asked in (marks)
11	2	PN junction	Ideal diode current	19-20 5(b) (8), 18-19 4(a) (7)
12	2	Metal-semiconductor contacts	Schottky and ohmic contacts	19-20 1(d) (6), 18-19 4(c) (4)
13	2	Excess carriers	Ambipolar transport	19-20 4(b) (7), 19-20 TT2 8(b) (7)
14	2	Excess carriers	Steady state	19-20 5(a) (7), 19-20 TT2 8(a) (6)

A sub-part with no marks after it had none printed on its paper.

12. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical	Diagram
Crystal structure	1	·	1	·
Quantum basics	7	·	6	·
Semiconductor in equilibrium	16	7	8	1
Carrier transport	3	4	5	·
Excess carriers	5	6	2	·
PN junction	4	2	5	1
Tunnel diode	2	·	·	·
Metal-semiconductor contacts	3	·	·	·
MOS structure	6	·	1	2
Bipolar transistor	7	·	2	·
JFET	·	·	2	·

13. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Semiconductor in equilibrium	130	23 %
2	Excess carriers	78	37 %
3	Carrier transport	75	50 %
4	PN junction	72	63 %
5	Quantum basics	58	73 %
6	MOS structure	54	83 %
7	Bipolar transistor	47	91 %
8	Metal-semiconductor contacts	16	94 %
9	JFET	13	96 %
10	Crystal structure	12	98 %
11	Tunnel diode	11	100 %

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